ECB makes a hawkish pivot at its March meeting

Rates have been kept on hold, but the European Central Bank has clearly opted for a hawkish pivot at today's meeting

ECB President Christine Lagarde at today's press conference in Frankfurt

Comments by European Central Bank President Christine Lagarde at the press conference gave today's meeting a more hawkish tilt. Even if a rate hike is not imminent, the change in tone and language acknowledges more uncertainty and is meant to demonstrate the ECB's willingness to act, if need be.

The fact that the well-known 'monitor closely' or 'closely monitoring' is back is a clear signal that the ECB has shifted to higher alert. In the past, the term 'monitor closely' had always been a sign of high alertness; the time it was used was during the short-lived banking tensions in March 2023 and before in 2022. In the distant past, 'monitor closely' was followed by 'vigilance' in the run-up to rate hikes. Following the logic of institutional ECB language, today's 'closely monitoring' combined with Lagarde's statement that risks to the inflation outlook were tilted to the upside are both clear signals of increased alertness.

New round of forecasts

Just for the record – though clearly not really relevant for future policy decisions despite a later cut-off date than usual (March 11) – the latest round of ECB staff projections has GDP growth coming in at 0.9% in 2026, 1.3% in 2027 and 1.4% in 2028. Inflation is expected to come in at 2.6%, 2.0% and 2.1% over the next few years.

In this base case scenario, ECB staff treats the current oil price shock mainly as a one-off – and one that would hardly necessitate a monetary policy reaction. However, in two alternative scenarios ECB staff prepared, the monetary policy reaction could be different as both scenarios worsen the stagflationary impact.

In the adverse scenario, the impact on the economy would be temporary and part of the inflationary impacts unwind. Growth would be some 0.3ppt lower than the baseline in 2026 and 0.1ppt lower in 2027. Inflation would be 0.9ppt higher in 2026 but would come down quickly, pushing headline inflation 0.5ppt lower in 2028.

In the severe scenario, energy prices would have a stronger and longer-lasting effect, reducing GDP growth by 0.5ppt and 0.4ppt in 2026 and 2027 respectively. The eurozone economy would be in a technical recession in the summer of 2026. Inflation would also be much higher, e.g., 3ppt higher than the baseline in 2027.

Decoding the ECB's reaction function

We reflected on the ECB's reaction function to oil price shocks and how it has changed over time here. During the press conference, Lagarde kept it vague, but reading between the lines suggested that the ECB is once again making a distinction between a supply-side and a demand-side shock. Stressing that the central bank would focus on commodity markets, potential supply bottlenecks, selling price expectations of firms, demand indicators and wage trackers, Lagarde gave the impression that the ECB would only start to react with rate hikes if and when higher energy prices and headline inflation would start to be passed through.

This is probably the most important lesson from the 2022 period: not that the ECB was too late to react to an energy price shock, but that it was too late to identify and to react to supply-driven inflation broadening to a fully-fledged and broad inflation problem.

On hold for now but rate hikes cannot be excluded

Looking ahead, an inflation wave is clearly in the making, from the direct impact of higher gas and oil prices on gasoline prices or retail energy prices next winter, to knock-on effects on transportation services, fertilisers and food prices. However, even a somewhat longer-lasting inflation overshoot does not have to be a concern. And, in fact, the only hurdle to calling this a typical supply side shock is the ECB's own institutional memory, its credibility as an inflation fighter and the fact that 'team transitory' turned out to be very wrong in 2022. It's obvious that 'transitory' has become a forbidden word at the ECB.

In a scenario in which the war in the Middle East and soaring energy prices remain limited in time, the ECB will talk like a hawk but not walk like a hawk – or, rather, fly like one. However, if energy prices stay high or higher for longer and find their way into other parts of the eurozone economy, the central bank apparently wouldn't shy away from rate hikes. Even though the bar for a rate hike seems to be higher than markets have currently priced in, given that the starting position for the ECB is currently very different from in 2022 (i.e., interest rates are higher), inflation and inflation expectations are much lower than in 2022.

All in all, a rate hike is not yet on the table, but today's meeting clearly marks a hawkish pivot.

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Disclaimer

ING's updated forecasts amid energy disruption

Our economists and strategists have updated their forecasts to reflect the rapidly changing events in the Middle East and our new base case scenario

Oil tankers and cargo ships are being forced to reroute as the Strait of Hormuz is effectively blocked

Interim forecast update

The war in the Middle East continues, and the Strait of Hormuz is practically blocked. Our initial base case assumptions from our March Monthly regarding the war have become outdated. We are normally not in the business of permanently changing our forecasts, but given that our next Monthly Update is only due in April, and we think that these times currently require some guidance, we have updated our full set of forecasts.

Three new scenarios

We have developed three new scenarios to guide our thinking and macro and market forecasts. In summary:

- **Scenario 1** (our new base case): Intensive combat will end within the next two weeks, but will be followed by lower intensity strikes that linger for several months. It's a scenario in which the opening of the Strait of Hormuz will take much longer than initially anticipated.

The 'end game' in this scenario could be a temporary ceasefire or tacit stand down with maritime escorts and exploratory talks on sanctions. Regime change over time is still possible, driven from the inside, as in East Germany in the late 1980s.

- **Scenario 2** (optimistic alternative): War ends surprisingly earlier than in our new base case scenario and the Strait of Hormuz opens up within a short period.
- **Scenario 3** (longer war): Longer combat action followed by protracted, lower-grade, impulsive confrontation. That would imply a much more uncertain situation in the Strait. In this scenario, the 'end game' could be a hard won ceasefire with third party guarantees (likely involving Russia or China), the possibility of sequenced sanctions relief, and a maritime security regime for Hormuz. However, even then, a durable peace could remain elusive, and the region might settle into a fragile, flare up prone equilibrium.

In short, our base case scenario does not foresee structural damage to an oil or gas production facility. We are facing a delay in energy and other shipments, but not (yet) a structural reduction. Needless to say, if a delay lasts long enough, it can still lead to these structural reductions. Generally speaking and for the time being, this is still mainly a supply-side shock, pushing up inflation and posing new challenges for central banks. As a result, the upward revision to our inflation forecasts is clearly larger than the downward revisions of growth, and they also differ across regions. What these new scenarios mean for our market forecasts can also be found here.

Contact us directly to discuss more details of the underlying scenarios and their implications.

ING Global Forecasts

	1Q26F	2Q26F	2026 3Q26F	4Q26F	2026F	1Q27F	2Q27F	2027 3Q27F	4Q27F	2027F
United States										
GDP (% QoQ, ann)	3.1	2.4	1.8	1.9	**2.5**	1.9	2.0	2.0	2.0	**2.0**
CPI headline (% YoY, avg)	2.5	3.8	3.5	3.2	**3.3**	2.8	2.0	1.9	2.1	**2.2**
Federal funds (%, eop)	3.75	3.75	3.50	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month SOFR rate (%, eop)	3.70	3.70	3.50	3.30	**3.30**	3.30	3.30	3.30	3.30	**3.30**
10-year interest rate (%, eop)	4.20	4.30	4.20	4.15	**4.15**	4.15	4.25	4.25	4.30	**4.30**
Fiscal balance (% of GDP)					**-6**					**-6.1**
Gross public debt / GDP					**102.6**					**104.5**
Eurozone										
GDP (% QoQ, ann)	0.7	0.6	1.0	1.5	**0.9**	1.6	1.6	1.5	1.4	**1.4**
CPI headline (% YoY, avg)	2.0	3.2	3.1	2.7	**2.8**	2.6	1.9	1.8	2.0	**2.1**
ECB Deposit Rate (%, eop)	2.00	2.00	2.00	2.00	**2.00**	2.00	2.00	2.00	2.00	**2.00**
3-month interest rate (%, eop)	2.00	2.00	2.10	2.10	**2.10**	2.10	2.10	2.10	2.10	**2.10**
10-year interest rate (%, eop)	3.00	3.00	2.95	2.85	**2.85**	2.90	3.00	3.00	3.10	**3.10**
Fiscal balance (% of GDP)					**-3.6**					**-3.5**
Gross public debt/GDP					**93**					**93.9**
Japan										
GDP (% QoQ, ann)	1.2	0.8	1.2	1.6	**0.8**	0.4	1.2	0.8	0.8	**1.0**
CPI headline (% YoY, avg)	1.7	2.4	2.4	2.3	**2.2**	2.6	1.8	1.8	1.6	**2.0**
Target rate (%, eop)	0.75	1.00	1.00	1.00	**1.00**	1.25	1.25	1.25	1.50	**1.50**
3-month interest rate (%, eop)	1.20	1.20	1.20	1.35	**1.35**	1.50	1.50	1.70	1.75	**1.75**
10-year interest rate (%, eop)	2.25	2.50	2.50	2.60	**2.60**	2.75	2.75	2.85	2.85	**2.85**
Fiscal balance (% of GDP)					**-4.6**					**-5.0**
Gross public debt/GDP					**230**					**228**
China										
GDP (% YoY)	4.6	4.6	4.7	4.5	**4.6**	4.2	4.9	4.4	4.4	**4.5**
CPI headline (% YoY, avg)	0.9	1.1	1.4	1.3	**1.2**	1.6	1.1	1.1	1.1	**1.3**
7-day Reverse Repo Rate (% eop)	1.40	1.40	1.30	1.30	**1.30**	1.30	1.20	1.20	1.20	**1.20**
3M SHIBOR (% eop)	1.55	1.50	1.45	1.45	**1.45**	1.45	1.45	1.45	1.45	**1.45**
10-year T-bond yield (%, eop)	1.85	1.90	1.95	2.00	**2.00**	2.00	2.05	2.10	2.15	**2.15**
Fiscal balance (% of GDP)					**-5.3**					**-5.3**
Public debt (% of GDP), incl, local govt					**145**					**-**
United Kingdom										
GDP (% QoQ, ann)	1.3	0.9	-0.1	0.8	**0.6**	1.5	1.5	1.5	1.5	**1.1**
CPI headline (% YoY, avg)	3.0	2.4	3.5	3.4	**3.1**	3.2	2.9	1.9	2.1	**2.5**
BoE official bank rate (%, eop)	3.75	3.50	3.25	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month interest rate (%, eop)	3.50	3.20	3.20	3.20	**3.20**	3.20	3.20	3.20	3.20	**3.20**
10-year interest rate (%, eop)	4.70	4.60	4.50	4.30	**4.30**	4.30	4.40	4.40	4.40	**4.40**
Fiscal balance (% of GDP)					**3.6**					**3.1**
Public sector net debt (FY, %)					**96.1**					**96.4**
EUR/USD (eop)	1.15	1.16	1.18	1.20	**1.20**	1.20	1.20	1.20	1.20	**1.20**
USD/JPY (eop)	158	158	155	153	**153**	150	148	147	145	**145**
USD/CNY (eop)	6.90	6.87	6.82	6.80	**6.80**	6.78	6.75	6.75	6.70	**6.70**
EUR/GBP (eop)	0.87	0.88	0.89	0.90	**0.90**	0.90	0.90	0.90	0.90	**0.90**
ICE Brent - US$/bbl (average)	76	91	85	77	**82**	72	70	67	63	**68**
Dutch TTF - EUR/MWh (average)	39	50	38	38	**41**	34	28	27	30	**30**

Source: ING forecasts

GDP Forecasts

QoQ% Annualised (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	3.1	2.4	1.8	1.9	1.9	2.0	2.0	2.0	2.5	2.0
Japan	1.2	0.8	1.2	1.6	0.4	1.2	0.8	0.8	0.8	1.0
Germany	-0.3	1.1	1.8	2.2	1.8	2.6	2.5	2.5	0.8	2.2
France	0.6	0.4	0.4	0.6	1.2	1.4	1.6	1.2	0.8	1.0
UK	1.3	0.9	-0.1	0.8	1.5	1.5	1.5	1.5	0.6	1.1
Italy	0.3	0.4	0.9	1.3	0.9	0.6	0.8	0.6	0.7	0.8
Canada	1.8	1.5	1.4	1.6	2.2	2.0	2.1	2.0	1.1	1.9
Australia	2.3	2.2	2.3	2.2	2.5	2.5	2.5	2.5	2.3	2.5
Eurozone	0.7	0.6	1.0	1.5	1.6	1.6	1.5	1.4	0.9	1.4
Austria	1.0	0.6	1.2	1.8	1.8	1.6	1.6	1.4	0.8	1.6
Spain	1.9	1.1	2.0	1.5	2.1	2.1	2.0	2.0	2.1	1.9
Netherlands	0.9	0.9	1.5	1.6	1.4	1.5	1.1	1.0	1.4	1.4
Belgium	0.8	0.4	1.2	1.4	1.3	1.3	1.2	1.4	0.8	1.2
Greece	1.6	0.8	1.8	1.7	1.7	2.4	2.2	1.7	1.9	1.8
Portugal	1.7	1.3	1.6	1.7	2.0	2.0	2.0	2.0	2.1	1.9
Switzerland	1.2	0.6	0.8	1.2	1.2	1.6	1.6	1.2	0.5	1.2

Emerging Markets, (YoY% growth)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Bulgaria	3.1	2.8	2.7	2.5	2.5	2.5	2.5	2.7	2.8	2.5
Croatia	3.5	3.1	3.0	1.7	1.6	1.7	1.8	1.8	2.7	1.7
Czech Republic	2.5	2.8	2.7	2.8	2.8	2.8	2.7	2.7	2.7	2.7
Hungary	1.6	1.4	1.7	2.2	1.9	3.2	3.1	3.6	1.7	3.0
Poland	3.6	3.8	3.7	3.6	3.6	3.4	3.0	3.0	3.7	3.2
Romania	-0.7	-0.8	0.3	2.8	3.1	2.9	2.6	2.7	0.6	2.8
Turkey	3.2	2.8	3.6	3.8	4.4	5.5	5.2	4.9	3.4	5.0
Serbia	3.5	3.0	3.1	2.6	2.8	3.2	3.3	3.6	3.0	3.2
Azerbaijan	3.0	2.0	4.0	1.0	3.0	3.5	2.0	3.5	2.5	3.0
Kazakhstan	4.0	5.0	5.0	6.0	5.0	4.8	4.0	4.0	5.0	4.5
Russia	1.0	1.0	0.5	-0.5	0.0	0.5	1.0	1.5	0.5	0.8
Uzbekistan	6.5	6.3	6.0	7.0	6.5	5.5	6.0	6.0	6.5	6.0
Ukraine	2.1	2.6	2.5	2.8	3.2	3.4	3.5	3.5	2.5	3.4
China	4.6	4.6	4.7	4.5	4.2	4.9	4.4	4.4	4.6	4.5
India	6.5	6.6	6.8	7.0	6.5	6.5	6.5	6.5	6.7	6.5
Indonesia	4.8	4.9	5.0	5.0	5.0	5.0	5.0	5.0	4.9	5.0
Korea	2.4	2.4	1.5	2.3	1.8	1.8	1.9	1.8	2.2	1.8
Philippines	4.0	4.5	6.0	6.2	6.0	6.0	6.0	6.0	5.2	6.0
Singapore	4.5	4.0	3.0	2.0	1.8	1.8	1.8	1.8	3.4	1.8
Taiwan	10.2	7.4	6.9	3.1	4.0	4.6	5.0	5.1	6.7	4.7

Norway: Forecasts are mainland GDP
Source: ING estimates

CPI Forecasts

YoY% (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	2.5	3.8	3.5	3.2	2.8	2.0	1.9	2.1	3.3	2.2
Japan	1.7	2.4	2.4	2.3	2.6	1.8	1.8	1.6	2.2	2.0
Germany	2.2	3.2	3.2	3.0	3.0	2.2	2.1	2.1	2.9	2.3
France	1.1	2.4	2.2	2.3	1.9	1.1	0.8	1.5	2.0	1.3
UK	3.0	2.4	3.5	3.4	3.2	2.9	1.9	2.1	3.1	2.5
Italy	1.5	2.3	2.6	2.6	2.2	1.7	1.6	1.9	2.3	1.8
Canada	2.1	2.8	2.9	2.5	2.1	1.7	1.8	2.0	2.6	1.9
Australia	3.7	3.7	3.2	3.2	3.0	3.0	3.0	3.0	3.5	3.0
Eurozone	2.0	3.2	3.1	2.7	2.6	1.9	1.8	2.0	2.8	2.1
Austria	2.3	2.7	2.6	2.3	2.2	1.9	1.8	2.0	2.5	2.0
Spain	2.6	3.3	2.8	2.6	2.2	2.2	2.1	2.1	2.8	2.1
Netherlands	2.3	2.5	3.0	3.1	2.6	2.3	1.7	1.6	2.7	2.0
Belgium	1.8	3.4	3.3	3.0	2.8	2.0	1.9	1.9	3.0	2.2
Greece	3.1	3.5	3.6	3.1	2.7	2.0	2.2	2.2	3.3	2.3
Portugal	2.2	2.9	2.6	2.4	2.0	2.0	1.9	1.9	2.5	2.0
Switzerland	0.4	1.1	1.0	0.7	0.6	0.6	0.8	0.8	0.8	0.7
Bulgaria	3.3	3.7	2.3	2.5	2.4	2.3	2.5	2.6	3.0	2.5
Croatia	3.8	4.1	3.3	3.2	2.4	2.6	2.7	3.1	3.6	2.7
Czech Republic	1.5	1.8	1.6	2.1	2.5	2.3	2.3	2.3	1.7	2.3
Hungary	1.9	3.4	3.6	3.9	4.0	4.2	3.8	3.4	3.2	3.8
Poland	2.5	3.6	3.7	3.3	3.1	2.2	2.0	2.3	3.2	2.4
Romania	9.7	10.4	6.0	5.4	4.2	4.1	3.9	3.7	7.9	4.0
Turkey	31.0	28.6	26.3	25.5	21.6	20.7	19.0	18.3	28.1	20.3
Serbia	2.6	3.0	2.6	3.4	3.3	3.5	3.8	4.2	3.0	3.8
Azerbaijan	5.8	5.5	4.9	4.6	4.3	4.5	4.7	4.9	5.2	4.6
Kazakhstan	11.8	11.0	10.3	10.4	10.0	9.3	8.6	8.1	10.9	9.0
Russia	5.9	5.4	4.8	5.3	4.5	4.8	5.3	4.7	5.3	4.8
Uzbekistan	7.4	7.1	7.0	7.7	8.0	7.7	7.6	7.6	7.3	7.7
Ukraine	9.2	8.5	7.5	7.0	6.5	6.3	6.1	5.7	8.1	6.2
China	0.9	1.1	1.4	1.3	1.6	1.1	1.1	1.1	1.2	1.3
India	3.0	3.7	4.1	4.3	4.5	4.8	4.8	5.0	3.8	5.0
Indonesia	3.0	2.3	2.5	2.5	2.5	2.5	3.0	3.0	2.6	2.5
Korea	2.1	3.0	2.5	2.1	1.8	1.3	1.9	2.2	2.4	1.8
Philippines	2.5	3.0	3.2	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Singapore	1.6	1.8	1.9	1.8	2.0	2.0	2.0	2.0	1.8	2.0
Taiwan	1.4	2.6	2.5	1.9	1.5	1.3	1.4	1.4	2.1	1.4

*Quarterly forecasts are quarterly average; yearly forecasts are average over the year, HICP for Eurozone economies
Source: ING estimates

Oil and Natural Gas Forecasts (avg)

	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Brent ($/bbl)	76	91	85	77	72	70	67	63	82	68
Dutch TTF (EUR/MWh)	39	50	38	38	34	28	27	30	41	30

Source: ING estimates

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Disclaimer

Assessing Europe’s Competitiveness as a Location for the Life Sciences Industry

Narrative report

efpia

CRA Charles River Associates

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Contents

Executive summary

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks further falling behind global peers, including China and the United States, which are outpacing Europe in attracting pharmaceutical investment, initiating more clinical trials and originating a greater share of novel pharmaceutical innovations.

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences industry is one of Europe's most important strategic assets, delivering medical breakthroughs that are fundamental to Europe's health, economic stability, and security.

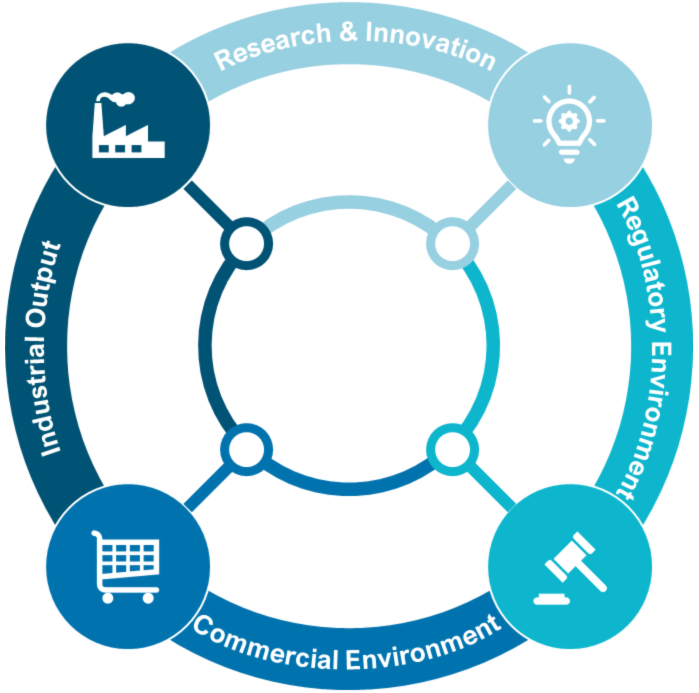
In the last two decades, Europe has experienced a decline in key drivers of competitiveness in the pharmaceutical sector. Without decisive actions to re-build a vibrant industrial ecosystem, this decline risks accelerating.

In practice, competitiveness in the life sciences sector is complex, shaped by a set of interconnected conditions. It is ultimately determined by how effectively it turns scientific capability into research and development (R&D) activities, which lead to approved products that reach patients and global markets at scale. The factors that drive life sciences competitiveness are many.

This report assesses the EU's life sciences competitiveness through this lens, focusing on four performance areas that most directly drive it: **research and innovation, the regulatory environment, the commercial environment, and industrial output.**

We compare the EU's attractiveness as a pharmaceutical investment location with that of global peers, including China, Switzerland, the United Kingdom (UK) and the United States (US).

Life sciences competitiveness is driven by four major performance areas.



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Overall, the EU's strengths lie in the quality of foundational research and industrial resilience. Still, challenges in regulatory speed, IP incentives, and commercial environment risk diminishing its global competitiveness relative to peers.

Research & Innovation

- The EU performs strongly in scientific excellence (second only to the UK) and digital capabilities (though it trails the US).
- The region trails comparators in industry R&D investment growth rate, falls behind Switzerland in R&D incentives, and faces a growing gap in clinical trial attractiveness compared to China and the US. The EEA share of global trial starts declined by 50% between 2013 and 2023.
- Patent applications have grown modestly (6% over 2014–2024), but China's 170% surge highlights the EU's relative stagnation.

Regulatory Environment

- The EU is lagging in approvals of new active substances (NAS), declining 20% over the past decade (vs. China's 470% increase)
- EU's minimal use of expedited pathways (0% in 2024 vs. 53% in the US) is a pressure point.
- Regulatory approval timelines have improved (430 days in 2024, down from 464 in 2015) but remain longer than in China (390 days) and the US (356 days).

Commercial Environment

- The EU's performance in spending on pharmaceuticals is moderate at 1% of its GDP, compared to China (1.8%) and the US (2.0%)
- In terms of inward FDI, the EU performs moderately with €1.5bn in 2023, compared to China (€2.3bn) and the US (€4.3bn).
- The EU performs weakly in the launch of NAS (39% of global launches 2012–2021 vs. 85% in the US).
- Mandatory industry refunds, such as high clawback rates (up to 53% in some EU countries) compared to 0% in China, Switzerland, and the US, deter investments.

Industrial Output

- The EU excels here, with strong manufacturing investment growth (15% compound annual growth rate (2018–2022), surpassing China's 11%) and a persistent trade surplus, reflecting resilient supply chains and export capabilities. However, maintaining this position cannot be taken for granted in an increasingly competitive global landscape.

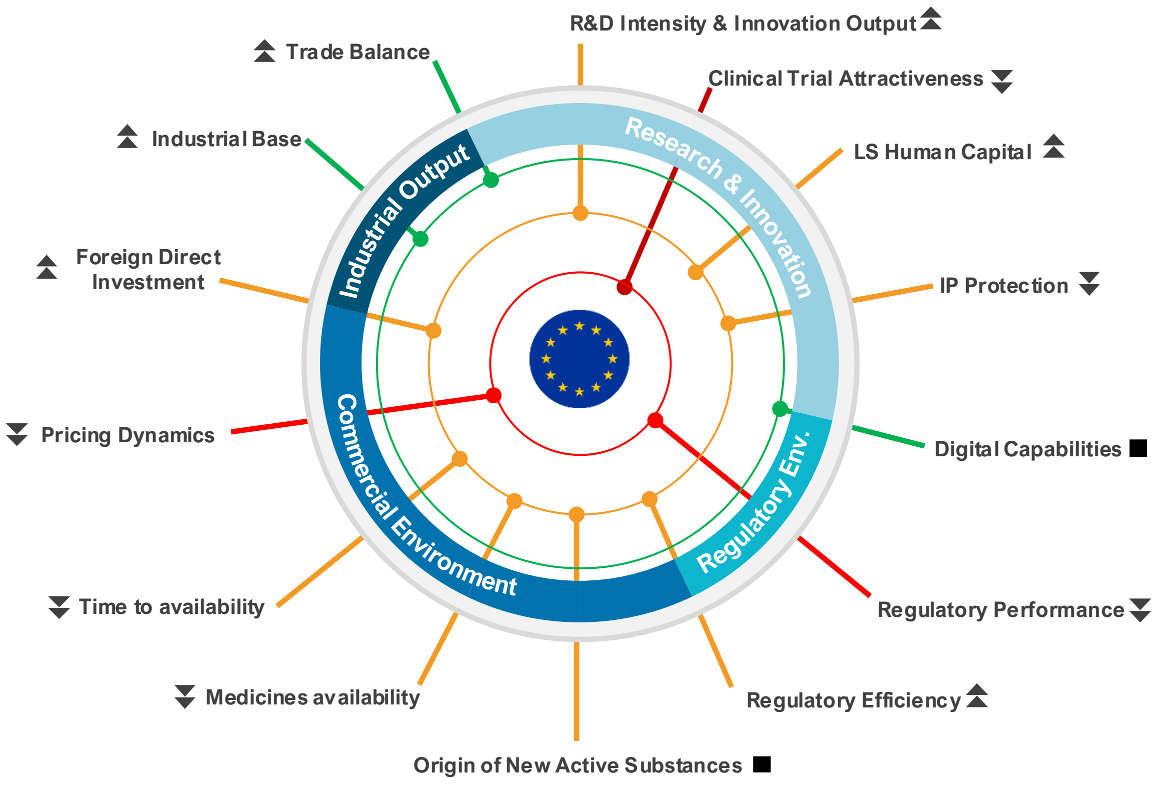
EU life sciences competitiveness at a glance

The EU retains deep scientific excellence, advanced digital capabilities, a strong industrial base, and high trade performance.

Yet, across the innovation pathway and uptake, the EU is ceding ground to the US and, increasingly, China – most visibly in the origin of new medicines, share of clinical trials, use of expedited regulatory pathways, as well as speed and breadth of medicines launches.

Europe's science and industrial base is an asset, but innovation conversion, as well as access breadth and speed, now define the EU's competitiveness gaps.

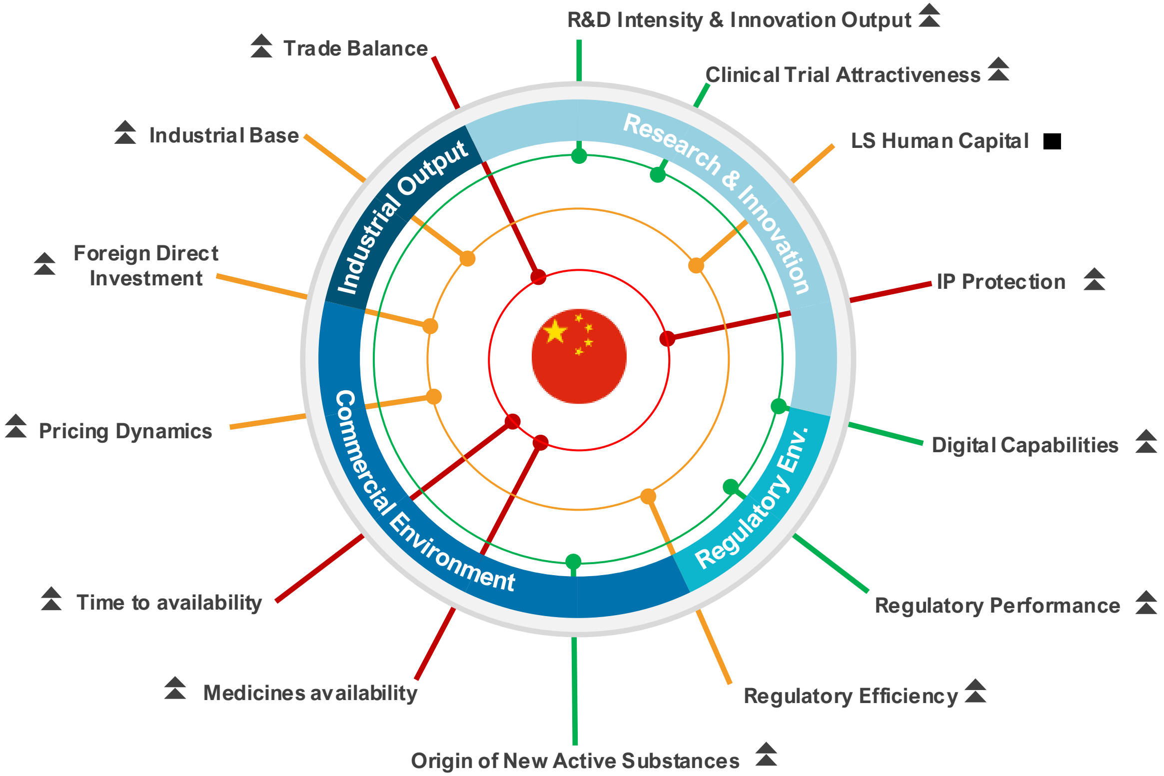
On those decisive metrics, the gap relative to the US persists, and the challenge from China is now unmistakable.

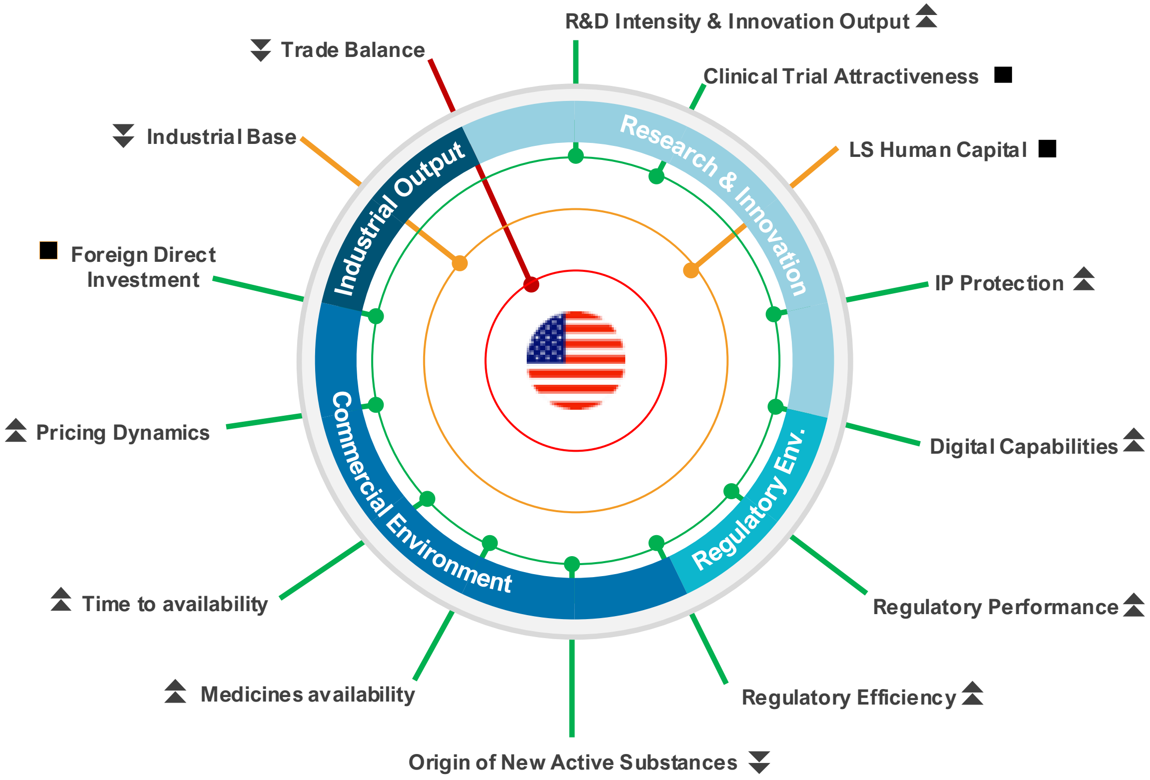


Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

China and US life sciences competitiveness at a glance





Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

The EU must act urgently over the next decade to strengthen its attractiveness relative to global leaders. Addressing weaknesses in key drivers of competitiveness could deliver significant gains in industry R&D investment, clinical trial activity, and the number of new active substances (NAS) originating in Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment

€105 billion

Additional industry R&D investment

- Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
- Growing at 8.5% CAGR, a rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.

Closing the gap on the share of global clinical trial starts

€17.9 billion

incremental total gross value added (GVA)

- To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.
- This would result in:
    - Nearly €18bn in the European economy.
    - 82,000 new jobs.
    - 158,000 more patients enrolled in clinical trials.

Closing the gap on the use of expedited review pathways

200 new medicines

approved via expedited reviews

- The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
- Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
- In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.

Closing the gap on originating NAS

100 medicines

originating from Europe

- China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
- Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Introduction

Why this report?

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences sector is one of Europe's most important strategic assets, delivering innovative medicines and vaccines that are fundamental to the long-term health and security of Europe.[1,2]

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks falling further behind global peers in research and innovation, where more ambitious, dedicated strategies are driving growth and outpacing Europe in attracting talent and cutting-edge developments.[1]

Europe stands at a pivotal crossroads amid a rapidly evolving global landscape. The acceleration of breakthroughs in biotechnology, personalised medicine, and AI-driven drug discovery is reshaping human health and economies alike. This, alongside intensifying geopolitical competition, supply-chain vulnerabilities, and widening gaps in scale-up capital, has raised a direct question for policymakers: can Europe convert scientific excellence into sustained leadership in innovation, investment, and patient impact?

In practice, competitiveness in the life sciences sector is a system outcome, shaped by a set of interconnected conditions: the ability to consistently generate breakthrough innovations, translate them efficiently into clinical and commercial products, manufacture and supply at scale, attract and retain talent and capital, and deliver timely patient access.

This report assesses Europe's life sciences competitiveness through that lens – focusing not only on research output, but on the full pathway from idea to impact. We compare the EU's attractiveness as a pharmaceutical investment location with that of four comparator countries: China, Switzerland, the United Kingdom and the United States. It draws on robust quantitative and qualitative analyses and insights to highlight where Europe is leading, where it is falling behind, and where targeted reform could unlock unrealised potential and improve Europe's competitiveness in life sciences.

This publication is expected to be updated periodically to support decision-makers who must balance multiple priorities to identify key policy interventions needed to realise Europe's competitiveness ambitions for its life sciences sector.

Comparator countries included in this analysis:

- European Union (EU)
- China (CN)
- Switzerland (CH)
- United Kingdom (UK)
- United States (US)

Performance analysis:

Using publicly available information, we assessed the latest EU performance relative to global comparators, as well as performance trend over 5 -10 years, where data allow.

What factors drive life sciences competitiveness?

The factors that drive life sciences competitiveness are many. Ultimately, the competitiveness of the life sciences sector is determined by how effectively it turns scientific capability into approved products that reach patients and global markets at scale. This report frames that pathway through four performance areas that most directly drive life sciences competitiveness: **research and innovation**, the **regulatory environment**, the **commercial environment**, and **industrial output**.

Assessment framework

A set of indicators and sub-indicators under each performance area was used to assess country/regional performance

Research & innovation	Regulatory environment	Commercial environment	Industrial output
R&D intensity	*Regulatory performance*	*Origin of new active substance*	*Industrial base*
Clinical trial attractiveness	*Regulatory efficiency*	*New medicine launch*	*Trade performance*
Life sciences human capital		*Time to market (launch)*	
Intellectual property protection		*Pricing dynamics*	
Digital capabilities		*Foreign direct investment*	

Chapter 1: Research and innovation

A conducive research ecosystem is the starting point. Countries and regions that lead in innovation are those that not only create an enabling research environment to generate new ideas but also sustain a high-throughput innovation engine that moves efficiently from proof of concept to scalable development programmes, with mechanisms and incentives that protect intellectual property.

We assessed Europe's R&D performance against comparator countries across key performance indicators:

Key performance indicators

R&D intensity
1. R&D investment growth
2. R&D investment levers
3. Scientific excellence
4. Patent applications

Clinical trial attractiveness
5. Number of global commercial clinical trial starts

Life sciences human capital
6. Life Sciences talent

Intellectual property protection
7. IP Index (IP protection strength)

Digital capabilities
8. Digital health maturity

1. Annual R&D investment growth

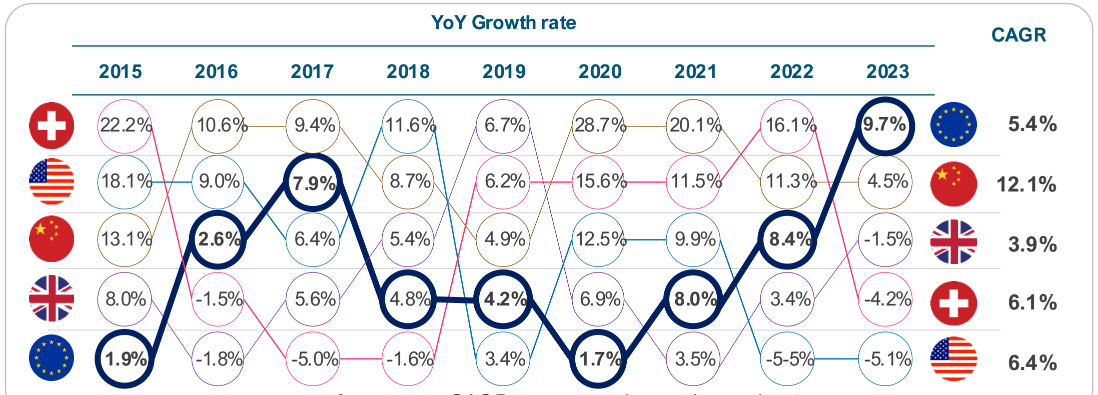
Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs weakly in growth rate of R&D investment from local and foreign companies, with growth rate about half of that in the US and two times lower than China.

Key performance indicator	Source	Latest
Growth rate (CAGR) of R&D investment by national and foreign pharma companies) [3-6]	EFPIA & ABPI	2023

1st	2nd	3rd	4th	5th
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



Acronyms: CAGR - compound annual growth rate

__NOTE:__ UK growth rates were derived from ABPI analysis of ONS 'business expenditure on research and development UK: 2024' available here. Data prior to 2022 is based on estimates following a methodology change in how UK BERD data is collected.

Over the last decade, European pharmaceutical R&D investment has grown steadily, but at a pace that may not fully match global shifts.[3-7] Annual growth rates in R&D investment by national and foreign pharmaceutical companies from 2015 to 2024 reveal distinct regional trends, as measured by the CAGR.

In EU countries, the CAGR is 5.4%, indicating steady growth. However, this pace falls behind growth rates in comparators. China exhibits the highest CAGR of 12.1%, followed by the US (6.4%) and Switzerland (6.1%). The lower growth rates in industry R&D investments in the EU highlight vulnerabilities in the EU's pharma ecosystem to attract investments.

The rapid acceleration observed in China, where investments surged notably in 2019 (28.7%) and 2020 (20.1%), reflects aggressive policy support for innovation and biotechnology. In absolute terms, the US is the global leader. R&D investments rose 65%, from €43bn in 2015 to €71bn in 2023, maintaining dominance, with expenditures nearly double those of the EU27.[3-6]

These trends have profound implications for the EU's pharma sector. Lagging growth relative to the US and China risks diminishing Europe's innovation leadership, potentially resulting in fewer breakthroughs in areas such as oncology and rare diseases.

While the EU's pharmaceutical R&D has grown reliably, sustaining global competitiveness cannot be taken for granted. In 2024, R&D investment growth in the EU was driven mainly by the health sector.[7] This demonstrates the sector's importance and the need for targeted policy measures to strengthen Europe's global standing and competitiveness.

1. Annual R&D investment growth

Strong EU performance

Moderate EU performance

Lower EU performance

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Key performance indicator	Source	Latest	
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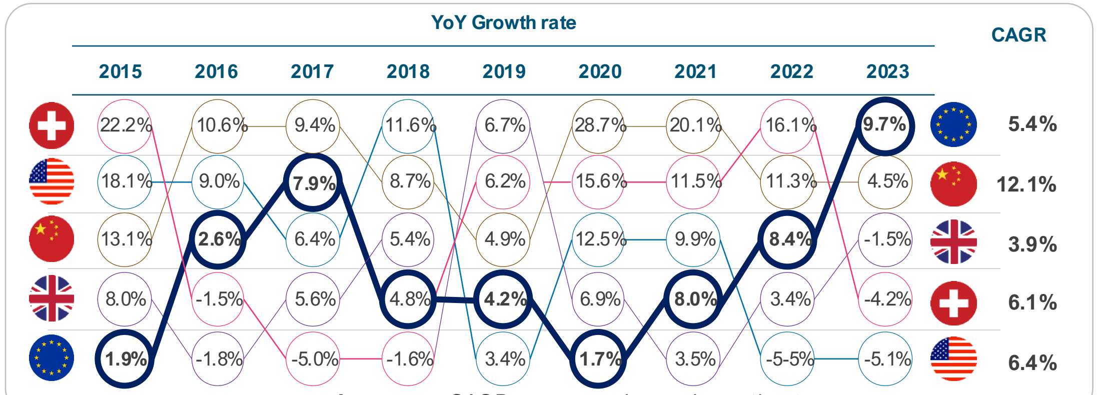
1st	*2nd*	*3rd*	*4th*	*5th*
China	US	Switzerland	EU	UK
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



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efpia CRA Charles River Associates

2. R&D investment levers

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately in the number and value of R&D investment levers. The EU lags Switzerland and the UK in the number of measures, and China and Switzerland in terms of their value.

Key performance indicator	Source	Latest
Implied subsidy rate on R&D expenditure (range) [8-12]*	EY & other sources	2023

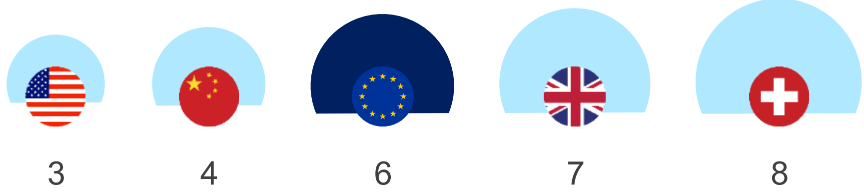
1st	2nd	3rd	4th	5th
22% - 32%	13% - 22%^	12% - 27%	1% - 7%	0% - 1%

**Simple benchmark comparison (large profitable firms). ^EU27 average*

***NOTE:** Implied or assumed subsidy rate can vary within a country/region. Switzerland has historically not operated a federal R&D tax credit. R&D expenses are simply deductible as normal business costs, resulting in a neutral tax treatment. Switzerland instead incentivizes commercialisation of IP, not the inputs to R&D*

There are 16 different types of R&D investment levers across countries [6]

Number of R&D investment levers available in comparator countries



R&D investment levers, such as tax credits and cash grants, are a core lever of competitiveness because they directly affect whether companies invest in long-term, high-risk research and where that investment is located.[12]

Based on the breadth of R&D support measures available, our analysis shows that the EU countries exhibit a relatively moderate level of availability, with an average of 6 instrument types per member state. [13] However, there is significant variation across the EU. This ranges from high performers such as the Netherlands (10) and Belgium (9) to lower performers like Romania (3).[13] Such diversity highlights the fragmented nature of R&D support in the EU. The EU's average (6) surpasses that of China (4) and the US (3), but trails Switzerland (8) and the UK (7).[13]

R&D tax credits or deductions are among the most commonly used investment levers across sectors as they play a decisive role in investment decisions. The value of R&D tax incentives (i.e., implied subsidy rate on R&D expenditure) widely varies across (and even within) countries. Data from OECD INNOTAX show that implied subsidy rate on R&D expenditures across the EU ranges from 0% to 39%.[20] R&D tax deductions typically range from 20% to 36% in high-signal EU member states, including France [14], Germany [15], and the Netherlands [16] In Switzerland, the R&D super-deduction (cantonal option) could be up to 50% [17], and up to 200% in China [18], making China and Switzerland more attractive for investment.

The EU, with an average implied subsidy rate of 16%, falls behind China (32%).[9-12,19,20,25] This, together with weakening EU incentive frameworks, particularly the reduction of the baseline RDP, may deter long-term investment. [27]

3. Scientific excellence

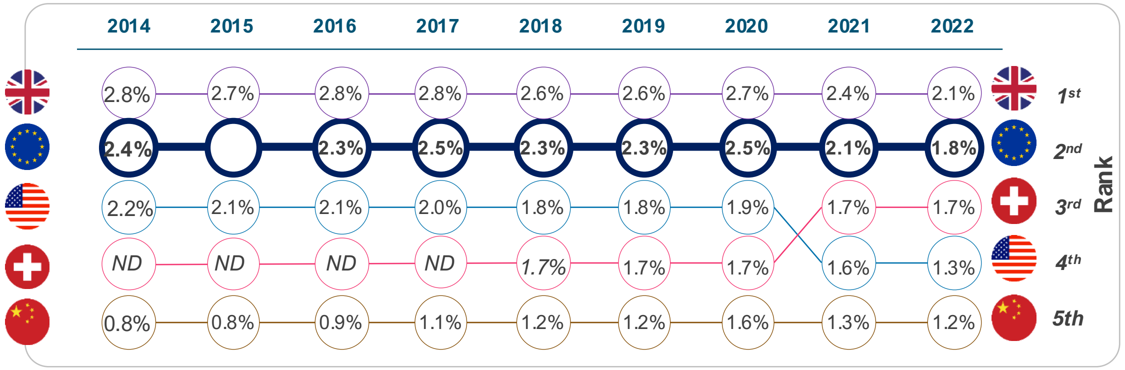
Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs strongly in the percentage of medical sciences publications which are amongst the most highly cited (top 1%) globally. Over the last decade, the EU has ranked only second to the UK on this performance indicator.

Key performance indicator	Source	Latest
Highly cited papers published (% of total) [28-30]*	Multiple sources	2024

1st	2nd	3rd	4th	5th
2.1%	1.8%^	1.7%	1.3%	1.2%

**Percentage of each country's medical sciences publications that are amongst the top 1% highly cited.*
^Average of France, Germany, and Italy



***Acronym:** ND – no data*

The proportion of a country's (or region's) medical sciences publications that rank among the most highly cited globally (top 1%) serves as a key bibliometric indicator of research quality, scientific excellence, and impact in the health sector.

This indicator is highly relevant to pharmaceutical competitiveness, as it underscores the strength of a country's foundational research ecosystem, which is essential for driving pharmaceutical innovation.

Relative to comparator countries, Europe's performance indicates resilience in research excellence, positioning it favourably against the UK and Switzerland—both renowned for high-impact outputs—and ahead of China and the US in terms of publication quality per output. [28-30]

Nonetheless, the trend over the last decade reveals a concerning reality: a 23% decline in performance in Europe between 2014 and 2022.

Although China lags comparators on this indicator, an interesting development is that it has increased by almost 50%, rising from 0.8% in 2014 to 1.2% in 2022. [30]

This demonstrates the scale of investment that drives China's transition toward scientific excellence and research impact.

If this trend continues, China is likely to overtake Europe soon, further eroding European competitiveness in research impact and scientific excellence.

4. Patent applications

Strong EU performance

Moderate EU performance

Lower EU performance

The EU performs moderately in terms of the number of patent applications relative to comparator countries. The share of global PCT applications from Europe has declined over the last decade.

Key performance indicator	Source	Latest
Patent (PCT) applications per capita [31]*	WIPO	2024

Rank	1st	2nd	3rd	4th	5th
Per capita	60.3	15.9	10.2	8.5	4.9
Total number	5,324	54,087	45,711	5,861	70,160

**This is the number of PCT applications per capita (across all sectors). Population data retrieved from the World Bank*

	2014	2017	2018	2019	2020	2021	2022	2023	2024		CAGR
	50.2	53.1	53.6	53.7	56.5	61.9	61.0	60.7	60.3		+2%
	19.3	17.4	17.2	17.6	17.9	17.9	17.7	16.4	15.9		-2%
	9.5	10.5	10.7	10.6	10.7	10.5	10.1	10.2	10.2		-1%
	8.2	8.4	8.5	8.6	8.8	8.6	8.4	8.1	8.5		+0.4%
	1.8	3.5	3.8	4.1	4.8	4.9	4.9	4.9	4.9		+10%

CAGR – compound annual growth rate

Patent activity is a critical driver of competitiveness, as it reflects a country's ability to generate, protect, and commercialise new technologies, and to attract investment in high-value, knowledge-intensive sectors.

The number of Patent Cooperation Treaty (PCT) applications filed by entities within a country or region, therefore, serves as a widely used indicator of innovation performance (not specific to pharmaceutical or biotechnology sectors). Analysis of annual PCT filings from 2014 to 2024 indicates a **slower trajectory for EU countries** compared with global peers. [32]

In absolute terms, China (70,160) ranked first in 2024. China (58,990) overtook both the US (57,840) and the EU (54,040) in 2019. On a per capita basis, Switzerland leads, with an average of 56 applications per 100,000 population per year, compared with 18 in the US, 10 in Europe, and 4 in China.[17] Over the period, the EU recorded a CAGR of -1% per year, below China (10% per year) and Switzerland (2% per year), showing a relative decline.[31]

Sector-specific data shows a similar pattern in pharmaceuticals. According to the European Patent Office (EPO), pharmaceutical patent filings declined by 13% in 2024, even as total European applications increased slightly by 0.3%. [32]

If sustained, this trend could gradually weaken the EU's pharmaceutical competitiveness and long-term innovation capacity.

5. Clinical trial attractiveness

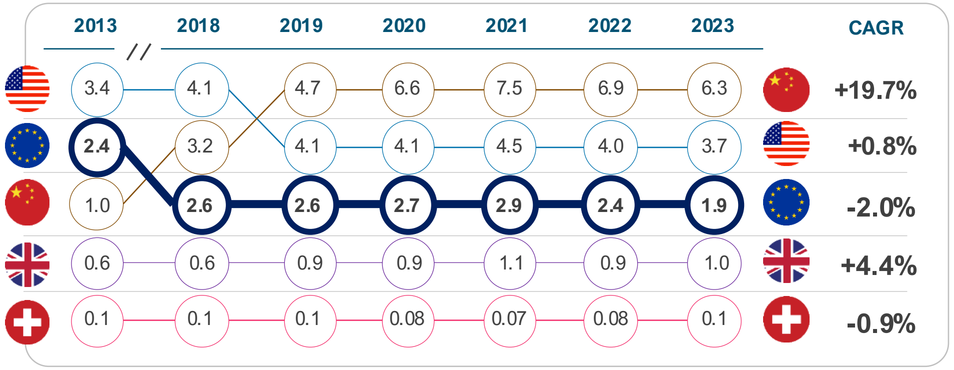
Strong EU performance | Moderate EU performance | Lower EU performance

Europe is losing ground in attracting clinical trials compared to its global peers. The EU's share of global industry clinical trial starts declined by 50% between 2013 and 2023.

Key performance indicator	Source	Latest
Number of global clinical trial starts [33]	IQVIA	2023

1st	2nd	3rd	4th	5th
			*	*
6,392	3,747	1,978	1,025	101

**This represents the number of commercial clinical trial starts only*
US data represent the North America region. EU data represent the EEA region



Number of clinical trial starts in '000 [33]

Clinical trial attractiveness refers to the extent to which a country or region is appealing to pharmaceutical sponsors for conducting clinical studies. This attractiveness is crucial for a country's competitiveness in the pharma sector, as it directly influences innovation, economic growth, and the country's global position within the industry.[3,33-35]

In the last decade, there has been a major shift in the geographic distribution of trials. In 2013, North America (including the US), the European Economic Area (EEA), and the rest of Europe (including Switzerland and the UK) accounted for 53% of global clinical trial starts (commercial and non-commercial). As of 2023, this figure stood at 33%. Over the same period, China's share increased from 8% to 29%, representing approximately 20% annual growth.[34,35]

Similarly, analysis of global commercial trial starts shows a similar trend. While EEA countries (except Spain) are losing share of global commercial trial starts, declining from 22% in 2013 to 12% in 2023, China continues to gain share, growing from 5% to 18% in that period.[34]

The rapid growth seen in China is not entirely surprising given its modernising society and market size. It is the worsening loss of market share in the EU that is concerning. Structural challenges, such as fragmented ecosystem and longer clinical trial start-up timelines have prevented the EU from growing to the level of opportunity that exists within the EU itself.[34] To strengthen its position as a global innovation leader, the EU should prioritise a more consistent implementation of the EU Clinical Trials Regulation and streamline processes for conducting multi-country trials.

6. Life sciences talent

Strong EU performance

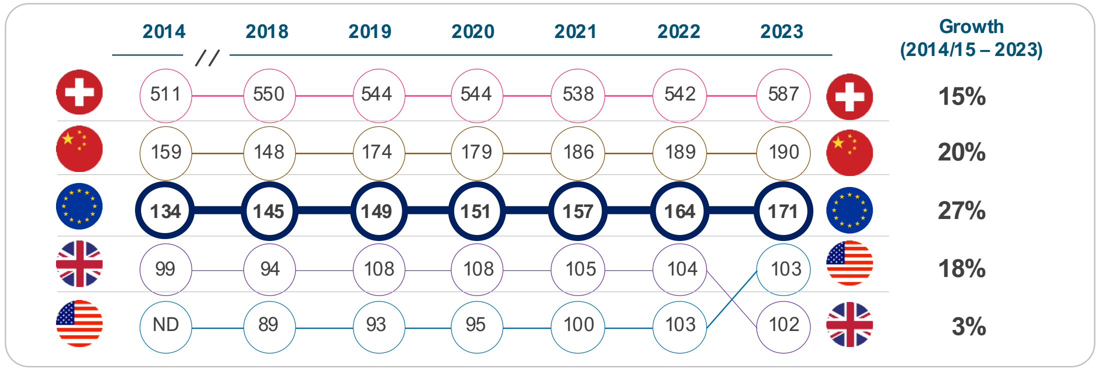
Moderate EU performance

Lower EU performance

Compared with comparator countries, EU countries perform moderately in human capital in the life sciences, as measured by the number of employees per capita in the pharmaceutical sector.

Key performance indicator	Source	Latest
Number of life sciences employees per capita (100,000 population) [3-7,36-37]	Multiple sources	2023

1st	2nd	3rd	4th	5th
587	190	171^	103	102



Percentage of graduates with statistics, maths, or natural sciences degrees (2023) [28]:

8.3 | 7.9 | 6.3 | 5.2^

^Aggregate of EU27 countries. ND – No data

Stock of human capital is an important driver of competitiveness in any sector.[38] A robust employment base fosters innovation by concentrating skilled professionals, such as scientists, engineers, and clinicians, who drive advancements in drug discovery, development, and manufacturing. Countries with high employment density in the life sciences sector are likely to attract higher levels of foreign investment if other competitiveness levers are in place.[38,39]

The EU27 demonstrated strong growth of 27%, rising from 134 to 171 employees per 100,000 population. This performance reflects collective advances across the EU in research, scientific excellence, and innovation, aimed at enhancing the sector's attractiveness.

In comparison, China achieved notable growth of 20%, elevating from 159 to 190, driven by substantial investments in biotechnology and pharmaceutical infrastructure amid its economic expansion.

Switzerland, already a leader in density, grew by 15% from 511 to 587, maintaining its position as a global hub for pharmaceutical giants and high-value research and development. The country's dynamism has been partly attributed to a powerful combination of world-class academic research, strong commitment to R&D, and a symbiotic relationship between academia and industry.[39]

Lastly, in terms of talent pipeline (graduates with natural sciences degrees), the EU falls behind comparators, with 5.2% compared to the US (8.3%).[40]

7. Intellectual property protection

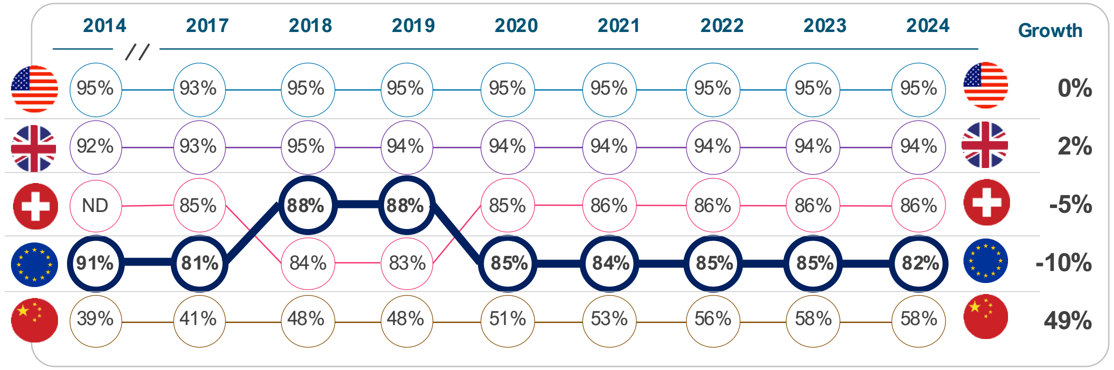
Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Relative to comparators, the EU is a moderate performer in intellectual property protection, only better than China.

Key performance indicator	Source	Latest
IP Index (IP protection strength) [41]	USCC	2024

1st	2nd	3rd	4th	5th
95%	94%	86%	82%^	58%

^EU average (unweighted)



ND – no data

The U.S. Chamber of Commerce's International Intellectual Property (IP) Index provides a comprehensive annual evaluation of IP frameworks across global economies, assessing factors such as patent rights, copyright protections, enforcement mechanisms, and adherence to treaties.[41]

The EU demonstrates a robust IP framework, consistently achieving average scores in the 73–88% range from 2014 to 2024, but performs below its peers, falling behind the US, UK, and Switzerland.[41]

IP systems in the US, UK, and Switzerland are generally characterised by stronger enforcement, fewer limitations on patentability, and more comprehensive incentives to support innovation.[41]

In the pharmaceutical sector, the EU's IP competitiveness has remained relatively stable. However, in recent years, it has shown signs of decline (-10%), partly owing to harmonisation reforms that introduce uncertainties in regulatory data protection and supplementary protection certificates.[41,42]

While China lags other comparator countries in IP protection strength, its upward trajectory (growing 49% over the last decade) signals China's growing emphasis on IP as part of broader industrial strategies, potentially narrowing the gap with global peers if current trends continue.[27,41-43]

In sum, the EU's pharmaceutical IP regime is less competitive than many of its peers and is already signalling a decline. Maintaining robust IP protections is essential to strengthening its competitiveness.

8. Digital health maturity

Strong EU performance | Moderate EU performance | Lower EU performance

Similar to comparator countries, the EU is a strong performer in digital health maturity. The available dataset for this indicator is immature and warrants monitoring in the future.

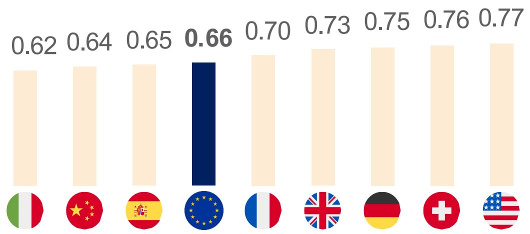
Key performance indicator	Source	Latest	EU
Digital maturity index [44]	WHO	2023	Strong

Most EU countries and the four comparator countries are ranked at being at the highest phase of digital health maturity by the Global Digital Health Monitor in 2023

The highest level of maturity is Phase 5, which signifies fully implemented, optimised, sustainable, and institutionalised systems that encompass advanced data governance structures and security, privacy, and device regulation, as well as robust infrastructure and scaled digital services.

Digital health maturity (GDHM 2023) [44]

AI Preparedness Index – AI Infrastructure (IMF 2025) [45]



Digital health, including data and artificial intelligence (AI) capabilities, continues to revolutionise healthcare delivery. Pharmaceutical companies are leveraging digital tools and AI capabilities to enhance efficiency in R&D execution and business planning. This indicates that countries with these capabilities are likely to increasingly become competitive.

The Global Digital Health Monitor (GDHM) provides a recent assessment of digital health maturity across a set of pillars not limited to digital/data governance, investment, and infrastructure, among other indicators. In this index, countries are rated on a scale from Phase 1 to 5 for each pillar, based on their level of maturity. The latest index (2023) rated most EU countries (18 of 27) and comparator countries at Phase 5 maturity.[44]

In terms of AI infrastructure, the AI Preparedness Index (2025) developed by the International Monetary Fund ranks EU countries and comparators highly in AI capabilities, with the US (0.77) leading the way, ranking above other countries, such as Switzerland (0.76), the UK (0.73), China (0.64) and many EU countries, including Germany (0.75) and France (0.70).[45] The average across the EU is 0.66, much lower than in the US, indicating some competitive advantage for the US. Reports from countries such as Germany suggest a **lag in the digitalisation of pharmaceutical R&D** relative to the US. [46]

Other assessments of AI capabilities, such as AI commercialisation [47] and AI research ecosystem [48], show similar rankings, with the US on top. The US demonstrates advanced digital capabilities driven by an ecosystem in which digital and AI tools are deeply integrated into research and patient care.[49]

Chapter 2: Regulatory environment

The regulatory environment shapes both speed to market and investment confidence. Competitive systems combine rigorous standards with clarity, consistency, and modernised pathways suited to assess increasingly complex health technologies. A predictable, efficient, and innovation-friendly regulatory system enhances a country's competitiveness.

We assessed Europe's regulatory performance and efficiency against comparator countries across key performance indicators:

Key performance indicators

Regulatory performance

9. Number of new active substances (NAS) regulatory approvals
10. Proportion of NAS approved via expedited review pathways

Regulatory efficiency

11. Median regulatory approval timeline for NAS (days)

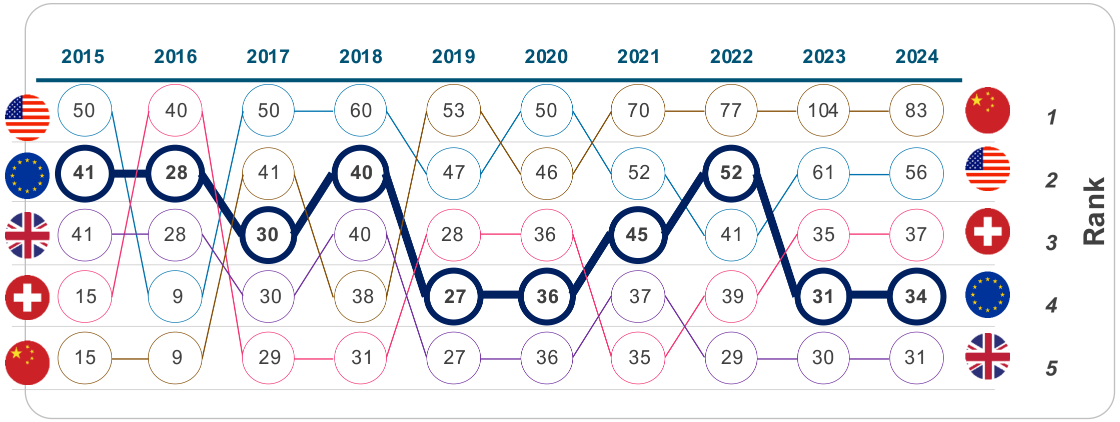
9. Regulatory performance – standard pathway

Strong EU performance | Moderate EU performance | Lower EU performance

EU regulatory output has grown more slowly than key global comparators over the past decade. Over the same period, China's performance increased markedly, driven by a sharp rise in NAS approvals.

Key performance indicator	Source	Latest
Number of new active substances (NAS) regulatory approvals [50-57]	Multiple sources	2024

1st	*2nd*	*3rd*	*4th*	*5th*
83	56	37	34	31



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

The regulatory environment plays a pivotal role in determining a country's pharmaceutical competitiveness.

Jurisdictions with streamlined regulatory approval systems that facilitate timely market entry for pharmaceutical innovations, which in turn accelerates patient access and reduces development costs, are likely to attract greater private investments, thereby enhancing competitiveness.

Number of new active substance (NAS) approvals is a good indicator of regulatory performance. Analysis of NAS approvals between 2015 and 2024 indicates that the EU's regulatory output has grown more slowly than that of some global peers over the past decade. Over this period, the EU recorded a relative decline of around 20% in NAS approvals compared with global comparators. [50-52]

Over the same period, China's performance improved by over 470%, from 15 approvals in 2015 to 83 in 2024, with a peak of 104 in 2023. [50-52, 56-61]

These trends do not reflect a lack of regulatory quality in the EU, but rather point to differences in system scale, resourcing, and the adoption of bold regulatory reforms in China.

As global R&D activity continues to expand, ensuring that the EU regulatory framework is adequately resourced and continuously modernised will be essential to support timely assessments, uphold high standards, and sustain the EU's role in global health standard-setting.

10. Regulatory performance – expedited reviews

Strong EU performance

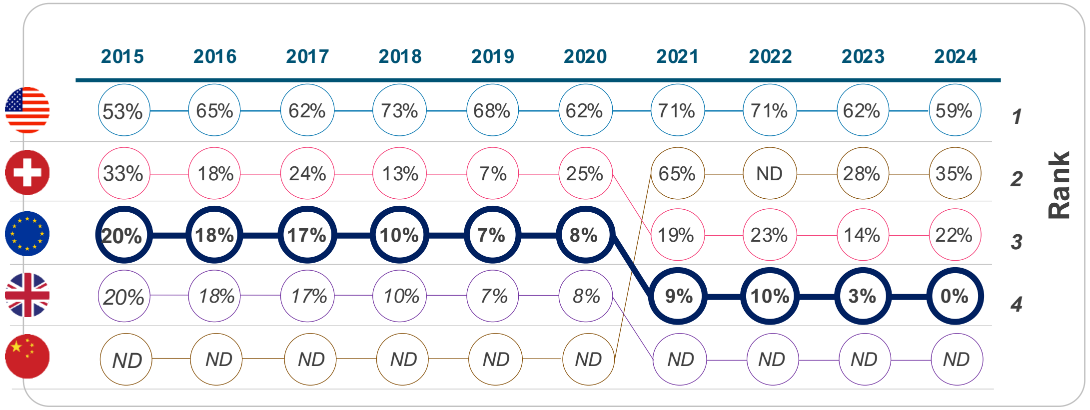
Moderate EU performance

Lower EU performance

Use of expedited review pathways differs across regions, with low uptake in the EU compared with global peers.

Key performance indicator	Source	Latest
Proportion of NAS approved via expedited review pathways [50-52, 56-57]	Multiple sources	2024

1st	2nd	3rd	4th	5th
59%	35%	22%	0%	*ND*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020). ND – no data

Expedited review pathways refer to ‘Accelerated Assessment’ by EMA (EU), ‘Fast Track’ by Swissmedic (Switzerland), and ‘Priority Review’ by the FDA/NMPA (US/China). These pathways are designed to expedite the evaluation and approval of pharmaceutical innovations.

Empirical evidence demonstrates that the EU employs expedited regulatory pathways significantly less frequently than global peers.[50-52,58-62]. In the last decade, the utilisation of expedited review mechanisms for regulatory approval of NAS within the EU declined from 20% in 2015 to 0% in 2024. By comparison, in 2024, expedited pathways accounted for 35% of approvals in China and 53% in the US.[50-52,58-62]

China’s recent regulatory reforms have played a crucial role in creating a robust end-to-end regulatory framework that aligns medicine approval processes with global standards.[61,63] In the US, expedited regulatory pathway tools are better interconnected; i.e., once a medicine receives Breakthrough Designation, it often receives Priority Review, Rolling Review, and Accelerated Assessment. In addition, the US has recently launched the FDA Commissioner’s National Priority Voucher to further expedite the evaluation of new medicines aligned with critical US national health priorities.[64]

Whereas in the EU, expedited review is a complicated process. Sponsors must first apply for Priority Medicines Scheme designation, then reapply separately to receive Accelerated Assessment, and reapply again to receive Conditional Marketing Authorisation.[65] Simplifying procedures, while maintaining high regulatory standards, would strengthen the EU’s overall regulatory performance.

ND – no data

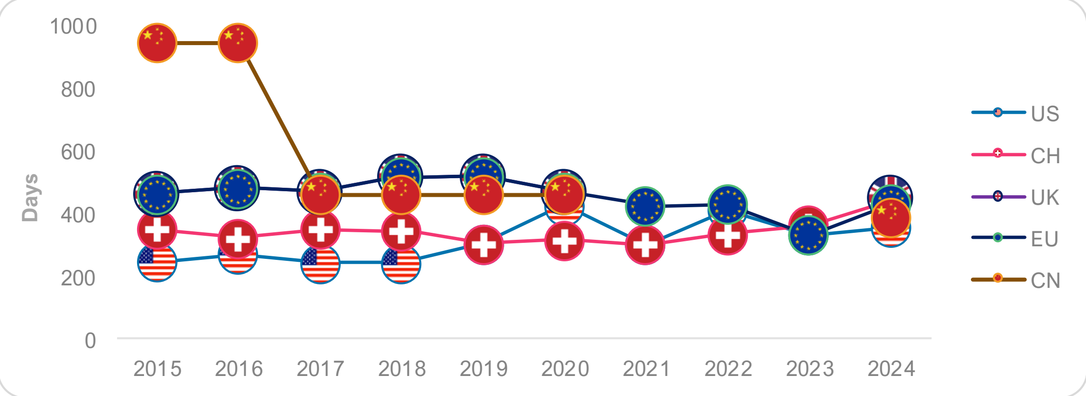
11. Regulatory efficiency – approval timeline

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Regulatory speed has improved in the EU over the last decade, performing moderately relative to peers in recent years.

Key performance indicator	Source	Latest
Median regulatory approval timeline for NAS (days) [50-52,66]	CIRS; ABPI	2024

1st	2nd	3rd	4th	5th
US	China	EU	Switzerland	UK
356	390	430	440	450*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

Regulatory speed is a key driver of competitiveness, as pharmaceutical sponsors are more likely to allocate resources to jurisdictions with predictable, expedited regulatory timelines.

Across jurisdictions, regulatory speed can be assessed using median approval timelines (from regulatory submission to regulatory approval).

Over the last decade, the EU's median approval timelines in the standard pathway declined, from 464 days in 2015 to 430 days in 2024, with a notable low of 333 days in 2023. [50-52]

However, the headline time-to-approval remains longer than that of China and the US. The median difference between China and the EU was ~40 days (430 vs 390 days), and ~74 days (430 vs 356) for the US vs the EU. Historically, the gap between the US and the EU has often been larger. [50-52]

While these trends suggest improvement in the EU's regulatory efficiency, the EU still lags behind global peers. The European Commission has proposed reducing standard assessment timelines to 180 days. This target is not yet a legal requirement until the legislation for the new EU Pharmaceutical package is formally adopted.[50-52]

Meeting this target will help narrow competitiveness and improve the EU's attractiveness to investors.

The commercial environment determines the extent to which innovation is funded and available to patients. Where commercial conditions are strong, companies often scale locally rather than relocating, and health systems adopt innovations without delay.

We assessed Europe's commercial environment performance against comparator countries across key performance indicators:

Key performance indicators

Origin of new active substances
- 12 Origin of regulatory-approved NAS (% of all NAS)

Medicine launch
- 13 Launch of new active substances (NAS) (% of all NAS)

Time to market (launch)
- 14 Median time from global launch to local launch (months)

Pricing dynamics
- 15 Cost-effectiveness thresholds for pricing and reimbursement
- 16 Pharmaceutical expenditure as a share of GDP
- 17 Mandatory industry refund rate on pharmaceutical revenues (%)

Foreign direct investment
- 18 Inward FDI in pharma (EUR)

12. Origin of new medicines

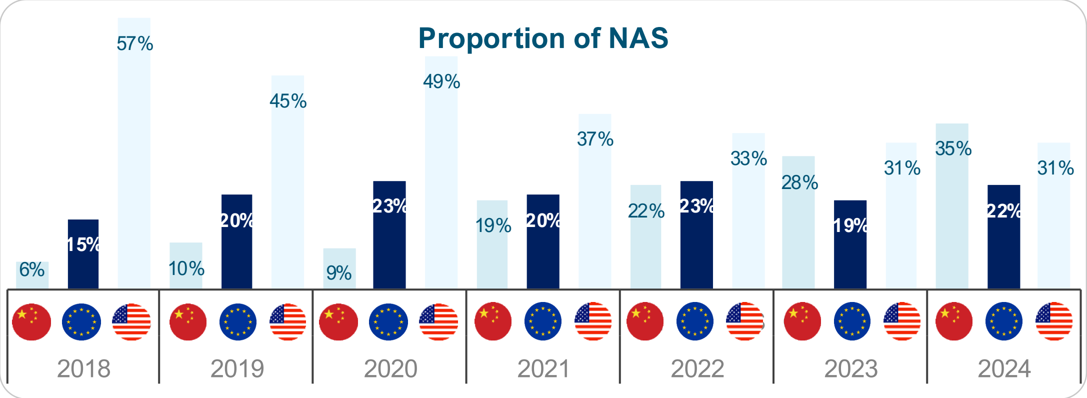
Strong EU performance | Moderate EU performance | Lower EU performance

Europe's performance in originating new active substances has remained broadly stable, while growth has accelerated in China and slowed in the United States.

Key performance indicator	Source	Latest	
Origin of regulatory-approved NAS (% of all NAS) [3-6]	EFPIA	2024	Moderate EU performance

1st	2nd	3rd
China	**United States**	**Europe***
35%	31%	22%

*This includes NAS originating from Switzerland and the UK as well.



In the early 2000s, Europe accounted for the largest proportion of globally originating NAS. Over time, the distribution has evolved, with China increasing its contribution and a more mixed trend in the US.

Europe's performance (including Switzerland and the UK) has remained relatively stagnant from 2021 to 2024, fluctuating between 17 and 19 NAS annually. Data show that EU-headquartered firms launched only ~130 NAS in 2015–24, compared with ~257 by US firms, a stark difference in innovative output. [3-6] Europe's trajectory differs from that of the US, which has continued to originate a high number of NAS, averaging 24 to 35 per year. This performance is associated with its large, unified market and supportive ecosystem – including strong venture capital funding, a favourable reimbursement environment, fast regulatory pathways, and substantial public and private R&D investment.[67]

China has also recorded rapid growth in NAS origination in recent years, overtaking Europe in 2023 and exceeding both Europe and the US in 2024. Output increased from 4 NAS in 2018 to 28 in 2024. Over the period 2018–2024, China's growth rate exceeded that observed in Europe, reflecting sustained investment in research infrastructure, government support measures, improvements in IP protection, and the expansion of domestic biopharmaceutical capabilities. [3-6, 67,68] Furthermore, NASs originating from China are increasingly being launched in both the US and Europe.[56]

While Europe continues to originate a stable number of new medicines, strengthening the conditions that support innovation will be important to maintain its role in an increasingly competitive global landscape.

13. Medicine launch

Strong EU performance

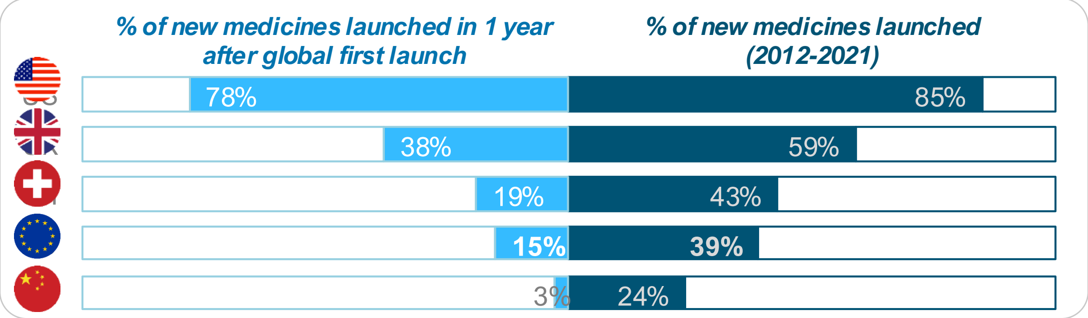
Moderate EU performance

Lower EU performance

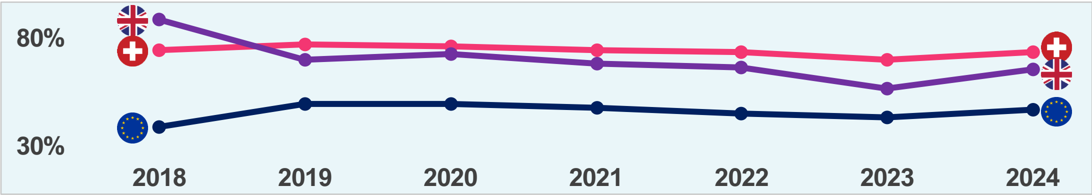
The EU's performance is moderate in terms of the launch of new medicines, compared to the US and the UK.

Key performance indicator	Source	Latest
Launch of new active substances (NAS) (% of all NAS) [69]	PhRMA	2021

1st	2nd	3rd	4th	5th
US	UK	Switzerland	EU	China
85%	59%	43%	39%	24%



Rate of availability (%) of NAS in Europe (EFPIA W.A.I.T.) [70]



The launch of new medicines (i.e., new active substances) is a core indicator of pharmaceutical competitiveness because it captures how effectively a health system converts scientific innovation into commercial reality, patient access, and sustained investment.

An analysis of data on 460 new medicines launched globally between 2012 and 2021 shows that the EU lags all comparator countries except China.[69]

On average, EU countries achieved a launch rate of 39%, substantially lower than the US of 85%. When focusing on timeliness, the EU average for launches within one year of global first launch is 15%, compared with 78% in the US, 38% in the UK, and 19% in Switzerland.[69]

Disparities persist even among leading EU performers, such as Germany (61% launched overall and 44% within one year), compared to France (52% launched overall and 23% within one year). While Germany tops Europe, it remains well below US levels.[69] The EU's relative underperformance in launch speed jeopardises its competitiveness, capturing only 16% of NAS sales (2019 – 2023) in top markets compared to 67% in the US.[70,71]

A similar trend is observed in NAS availability (public reimbursement) **with wide variability within the EU**. In 2023, an average of 46% of NAS was reimbursed in the EU, lower than in Switzerland (73%) and the UK (65%).[70]

The EU's lagging performance in NAS launches and availability poses significant risks to pharmaceutical competitiveness, diminishing its role as a global innovation leader.[71]

14. Time to market (launch)

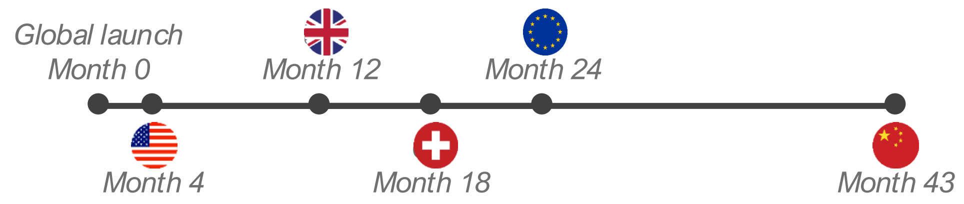
Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Compared with peers, the EU performs moderately on the time to local launch of new medicines (NAS).

Key performance indicator	Source	Latest	EU
Median time from global launch to local launch (months) [69]	PhRMA	2023	Moderate

1st	2nd	3rd	4th	5th
US	UK	Switzerland	EU	China

Average months from global first launch to local launch



The EU's performance on the time-to-launch of new medicines, measured as the average number of months between a new medicine's global first launch and its local launch, reveals notable disparities relative to global peers. The average delay in the EU stands at 24 months, though it widely varies across EU countries, **ranging from 128 days (4 months) in Germany to 840 days (28 months) in Portugal**. A similar trend is highlighted in EFPIA's W.A.I.T Indicator Survey.[70]

In comparison, the US demonstrates the shortest average delay at 4 months, enabling significantly faster access for patients. The UK and Switzerland follow with average of 12 and 18 months, respectively – both outperforming the broader European average. [69] The US consistently demonstrates superior performance in both the proportion of NAS made available and the speed of their launch, underscoring a regulatory and market environment conducive to rapid innovation deployment. The wide gap with the US underscores challenges to the EU's pharmaceutical competitiveness. Prolonged delays in the launch of new medicines may deter global investors from prioritising EU markets for initial launches, as seen in the US's dominance in global first introductions.

Fragmentation and slower access, with variations among member states, signal an unattractive environment. Such patterns disincentivise R&D investment within the EU, limit patient access, and erode the region's role as a hub for pharmaceutical innovation.[71]

To enhance competitiveness, the EU would benefit from policies that streamline and expedite regulatory approvals, strengthen IP protections, and promote reimbursement systems that truly reward innovation, aligning more closely with leading performers such as the US.

15. Cost-effectiveness thresholds

Strong EU performance | Moderate EU performance | Lower EU performance

The EU shows moderate performance in willingness to pay (WTP) for a standard unit of health benefit (QALY) gained from new medicines at an average cost per QALY of €43,500.

Key performance indicator	Source	Latest
Cost-effectiveness thresholds for pricing and reimbursement [72-75]	Multiple sources	2024

1st	2nd	3rd
High	Moderate	Low

China, Switzerland, and the US do not use CET. CET for Switzerland is implied based on a peer-reviewed source.

** Implied (peer-reviewed sources)*
^ Average cost-per-QALY threshold across EU27, for explicit and implied thresholds

A country's pharmaceutical cost-effectiveness threshold (CET) is a critical factor for investment attractiveness and competitiveness because it directly influences market access and pricing and reflects the value placed on innovation. A higher, or more flexible, CET can signal a more favourable market for innovative medicines, encouraging pharmaceutical companies to launch products and invest in research and development in that region. Conversely, low or strict thresholds lead to slower adoption of innovations and reduced investment.

China and the US do not use CET; therefore, they are excluded from the analysis. Switzerland's CET is implied from a peer-reviewed source, but the country does not officially use CET. [73-75]

A recent analysis shows that, across EU countries, the average cost per QALY threshold is €43,500 (~1.26 times the average GDP per capita), lower than Switzerland's (€100,000), though higher than the UK's (€28,500).[72] The UK's CET is set to increase to €35,000 from April 2026.[76]

The average CET across other high-income countries, including Australia, Canada, Japan, New Zealand, Norway, and South Korea, is €36,100.[72]

We note the increasing stringency of reimbursement criteria across the EU, which is impacting the breadth of availability of new medicines. For example, an analysis of HTA outcomes in France, Germany, the Netherlands, and Sweden shows a decrease in the number of positive HTA recommendations in 2023 relative to the averages from 2019 and 2022.[77] The increasing stringency of market access policies in Europe is widening the competitiveness gap.

16. Pharmaceutical expenditure

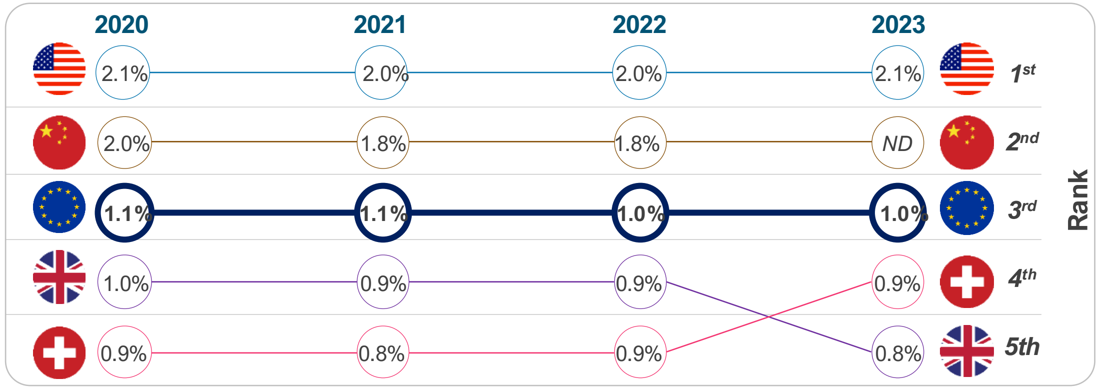
Strong EU performance | Moderate EU performance | Lower EU performance

Compared with the US and China, the EU, Switzerland, and the UK devote a lower share of GDP to medicines.

Key performance indicator	Source	Latest	
Public pharmaceutical expenditure as a share of GDP [78,79]	Multiple sources	2023	

1st	*2nd*	*3rd*	*4th*	*5th*
2.1%*	1.8%*	1.0%	0.9%	0.8%

**This represents total pharmaceutical expenditure. For China, the latest data available is for 2022.*



ND – no data

NOTE: ESTIMATES OF PHARMACEUTICAL EXPENDITURES WERE DERIVED FROM MULTIPLE SOURCES; THEREFORE, CAUTION MUST BE TAKEN WHEN MAKING COMPARISONS.

Pharmaceutical expenditure (expressed as a percentage of GDP) serves as a proxy for a health system's capacity and willingness ("market pull") to fund and utilise pharmaceutical innovations at scale. When market pull is high and predictable, it strengthens pharmaceutical competitiveness by improving the expected returns to innovation, accelerating launch of new medicines, and supporting an ecosystem (clinical infrastructure, evidence generation, specialised supply chains) that attracts further investment. [80,81]

Analysis of pharmaceutical spending in comparator countries shows that the US and China exhibit higher shares of GDP on pharmaceuticals, driven by innovation and population scale, respectively. Whereas European countries, including Switzerland and the UK, emphasise cost containment through rebates and procurement, keeping shares lower.[79-81]

Evidence shows that pharmaceutical innovation increases when companies can expect a larger market, as bigger markets attract more new medicines.[82]

New medicine launches are also responsive to expected prices and revenues, while price regulation and spillover mechanisms are associated with launch delays and a smaller number of new medicines developed.[82]

Countries that combine adequate pharmaceutical spending with predictable, value-based reimbursement and timely access tend to be more attractive launch and investment markets – strengthening pharmaceutical competitiveness through a virtuous cycle of uptake, evidence generation, and reinvestment.

17. Mandatory industry refunds*

Strong EU performance | Moderate EU performance | Lower EU performance

High use of mandatory industry refunds, such as clawbacks, negatively affects pharmaceutical investment in Europe.

Key performance indicator	Source	Latest
Mandatory industry refund rate on pharmaceutical revenues (%) [66,78,83-86]	Multiple sources	2024

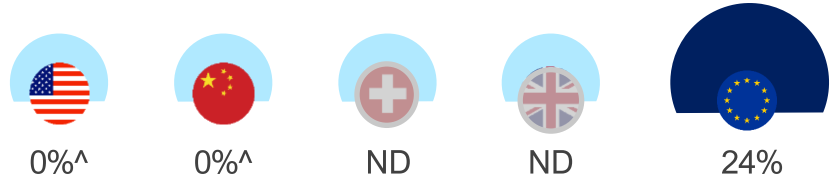
1st	2nd	3rd	ND	ND
US	China	EU	Switzerland	UK

**Mandatory industry refunds include mandatory rebates, mandatory sales tax, clawbacks (budget caps), and refunds under managed entry agreements.*

ND – no directly comparable data

Mandatory industry refund policies exist only in European countries [78,83-87]

Mandatory industry refunds as a percentage of public pharmaceutical revenue (2024)



^ There is limited or no mandatory industry refund requirement.
ND – no directly comparable data

In Europe, mandatory industry contributions, such as clawbacks, are increasingly being used to contain pharmaceutical spending, with little regard for their implications for Europe's innovation ecosystem and patient access.

Clawbacks are payments that companies are required to make back to public payers when pharmaceutical spending exceeds agreed budgets. In Europe, these payments are typically applied after the fact and are calculated as a percentage of sales or as a share of budget overspend.

Clawbacks are widely used across the EU as a cost-containment tool, with more than 20 Member States applying some form of clawback mechanism. In several countries – including Belgium, France, Greece, Italy, and Romania – clawback levels are particularly high. [66,78,83,87] For example, in 2024, industry clawbacks accounted for around 53% of total pharmaceutical revenues across all channels in Greece, while in France, industry payments increased from €72 million in 2019 to over €1.6bn in 2023. [83,87] Overall, mandatory industry refunds in Europe have risen from around 13% to 24%, growing 3 times faster than public payer spending since 2018.[78]

By contrast, comparator countries such as the US and China do not apply national clawback systems, relying instead on more limited rebates or discounts. [84-86]

Robust evidence indicates that extensive use of clawbacks (and other cost-containment mechanisms) negatively affects pharmaceutical investment, innovation, and patient access, with implications for Europe's long-term competitiveness in life sciences. [88]

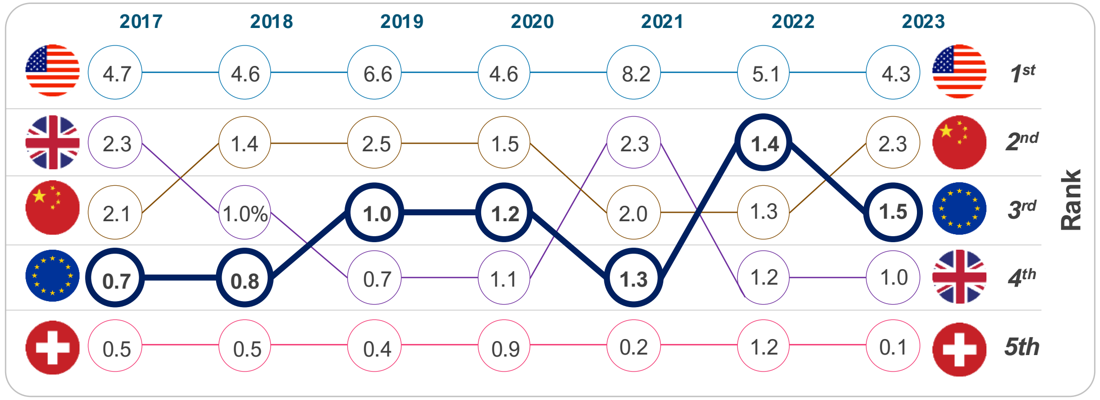
18. Inward FDI into pharmaceutical sector

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately relative to comparators in attracting inward FDI to the pharmaceutical sector.

Key performance indicator	Source	Latest
Inward FDI in pharma (EUR bn) [28,89]	EFPIA & OLS	2023

1st	2nd	3rd	4th	5th
4.3	2.3	1.5*	1.0	0.1



Inward foreign direct investment (FDI) refers to investment by foreign companies in domestic production, research facilities, and infrastructure. In the pharmaceutical sector, inward FDI matters because it supports clinical trials, research hubs, manufacturing capacity, and integration into global innovation networks, all of which contribute to long-term competitiveness.

In the EU, inward pharmaceutical FDI has increased steadily in recent years, rising from around €0.7bn in 2017 to €1.5bn in 2023.[89] However, the overall scale remains lower than in key comparator countries. The US continues to attract significantly higher levels of investment, reporting €4.3bn in a down year (2023), roughly 3 times the EU average. The 2021 spike (€8.2bn) indicates the US continues to attract transformational investments as firms rebase supply chains or build new platforms, as seen during the pandemic.[88]

China also remains a strong competitor for pharmaceutical capital, supported by market size, industrial policy, and scale advantages. In the first seven months of 2025, the biopharmaceutical sector accounted for a third of China's total inward announced FDI of approximately €10bn.[90]

The UK shows a declining trend since Brexit, while Switzerland's relatively low inflows reflect its role as a home base for global pharmaceutical companies rather than a lack of sector strength. [28,89,91].

Overall, the trend in the EU is moving in the right direction, but it still does not capture a proportional share of global mobile pharma capital. The challenge is not momentum; it is magnitude, particularly when compared with the performance of the US and China.

Chapter 4: Industrial output

Industrial output is a clear expression of competitive strength: the capacity to manufacture, supply, and export life sciences products at scale. This includes the depth of biomanufacturing capabilities, the maturity of supply chains, and trade policies to promote and sustain manufacturing leadership and export capacity.

We assessed Europe's industrial performance against comparator countries across key performance indicators:

Key performance indicators

Industry base

19 Pharmaceutical manufacturing investment growth (%)

Trade performance

20 Trade balance in pharmaceuticals

19. Industrial base

Strong EU performance

Moderate EU performance

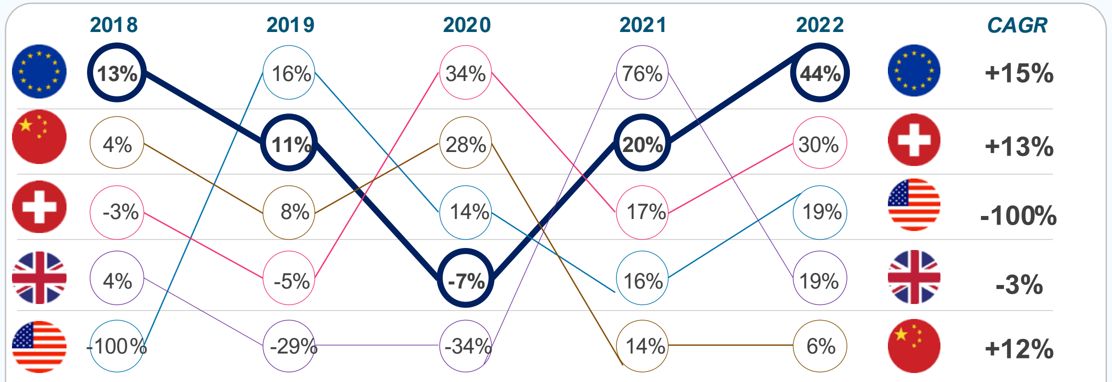
Lower EU performance

EU countries perform strongly relative to comparators on pharmaceutical manufacturing investment growth between 2018 and 2022.

Key performance indicator	Source	Latest
Pharmaceutical manufacturing investment growth rate [92,93]	UNIDO & NBS China	2022

1st	2nd	3rd	4th	5th
44%*	30%	19%	6%	-2%

This represents the unweighted average across EU countries.



CAGR – compound annual growth rate

We measured the growth rate of pharmaceutical manufacturing investment using the annual growth rate in Gross Fixed Capital Formation for pharmaceutical and medicinal-chemical manufacturing (ISIC 21) as a proxy (UNIDO data for the EU, Switzerland, the UK, and the US, and the National Bureau of Statistics for China). [92,93]

Between 2018 and 2022, EU countries demonstrated strong growth in pharmaceutical manufacturing investment, with a CAGR of 15%. This positioned the EU ahead of comparators.

China's performance, while steadily positive across all years and yielding a notable 12% CAGR, was outpaced by the EU. Compared with Switzerland, the EU's higher CAGR indicates a more consistent upward trajectory.

The UK's negative CAGR reflects pronounced instability, with sharp declines in 2019 (-29%) and 2020 (-34%) offsetting a temporary spike in 2021 (76%). The US experienced a severe contraction between 2017 and 2018, which may be connected to diminished domestic investment incentives at that time.[94,95]

These dynamics have significant implications for the EU's competitiveness in the global pharmaceutical sector. The EU's superior growth rate indicates a more expanded manufacturing infrastructure and greater resilience to external shocks, positioning it favourably relative to competitors.

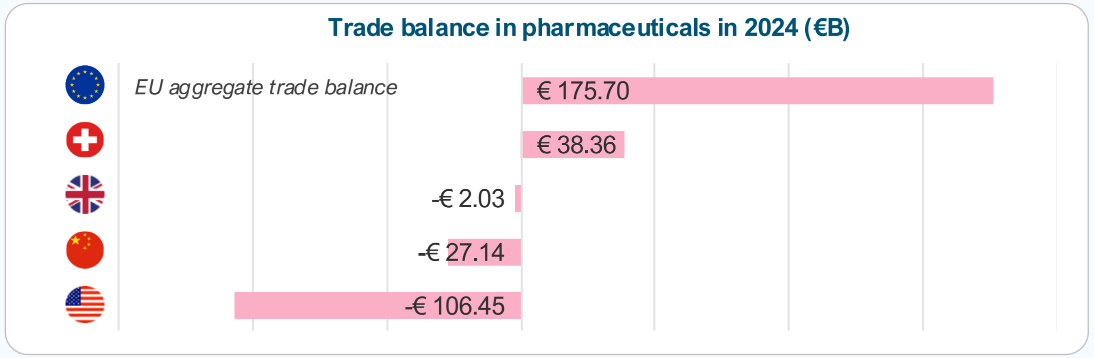
Overall, sustained investment growth bolsters the EU's strategic autonomy, fosters job creation, and enhances its attractiveness to multinational firms, thereby reinforcing long-term economic and health security advantages.

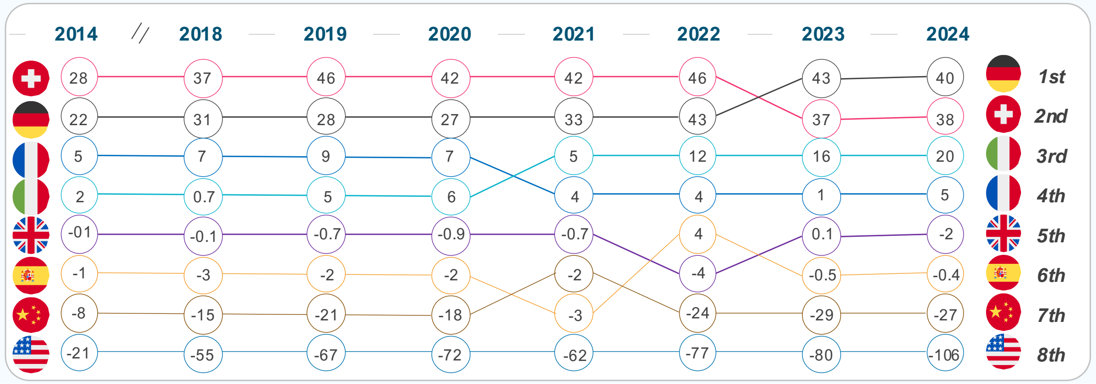
20. Trade performance

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

The EU's trade performance on pharmaceuticals is strong relative to comparators, with many top EU countries maintaining a positive trade balance in the last decade.

Key performance indicator	Source	Latest	
Trade balance in pharmaceuticals [96]	UN Comtrade	2024	





Trade balance in pharmaceuticals is a key indicator of the sector's global competitiveness. A positive trade balance implies several strategic advantages.

Analysis shows the EU has maintained a persistent and growing trade surplus, demonstrating its strength in exporting high-value pharmaceuticals and active pharmaceutical ingredients.[96,97] For instance, without the pharmaceutical industry, the EU trade balance would go from a surplus of €133bn to a €88bn deficit. Latest Eurostat data show that both EU pharmaceuticals exports (€366.5bn) and imports (€145.5bn) increased in 2025 compared to 2024. Exports increased more than imports, resulting in a trade surplus of around €221bn in 2025, up from around €194bn in 2024.[98] This surplus underscores the EU's ability to produce goods that meet international demand.

Compared to peers, EU exports of pharmaceuticals to the US amounted to €160.6bn in 2025 (compared to €121bn in 2024), while imports from the US in 2025 were €60.1bn (compared to €45.6bn in 2024). To China, pharmaceutical exports decreased from €16.9bn in 2024 to €13.9bn in 2025, while imports increased threefold, from €4.4bn in 2024 to €13.1bn in 2025, signaling an increase in China's domestic capacity to research and develop new medicines.[98]

While EU trade performance is strong, this should not be taken for granted. This trade performance must be contextualised amid challenges that could erode it, such as inefficient regulatory frameworks, which may necessitate reforms to streamline approvals and reduce bureaucratic hurdles. Proactive policies are needed to sustain trade performance against global pressures.[99]

Conclusion

This evidence-based assessment identified strengths and trends of the EU's life sciences ecosystem, together with key pressure points for improving competitiveness.

Strengths to build on

- **Scientific excellence & research base.** The medical research outputs of many EU countries remain high-quality according to bibliometric measures, providing a strong foundation for developing and scaling innovations.
- **Digital capabilities.** Digital capabilities are advanced, enabling platforms for digital health tools, AI-driven clinical trials, and data-driven regulation.
- **Industrial and trade performance.** In the EU, investments in pharma manufacturing continue to grow, while the pharmaceutical trade surplus remains, signalling robust manufacturing and export capacity. Yet sustaining this position cannot be taken for granted as global competition intensifies.

Priority pressure points for improving competitiveness

- **R&D investment levers.** Instruments are heterogeneous in breadth and value across the EU. Fragmentation and administrative burden reduce attractiveness relative to Switzerland and China.
- **Patents and capital formation.** Patent applications and inward pharma FDI trends indicate that the EU is not capturing a proportionate share of mobile innovation capital relative to global peers.
- **IP protection uncertainty.** The IP framework is less predictable than in the UK and the US.
- **Clinical trial attractiveness and innovation output.** Over the past decade, the EU's share of global trial starts has fallen to 12%, while China now accounts for ~30% of global trial starts.
- **Origin of new medicines.** China has overtaken both the EU and the US as NAS originator, increasing from 4 NAS in 2018 to 28 in 2024.
- **Regulatory framework.** Expedited pathways are underutilised compared to peers, while approval timelines, though improving, remain much longer.
- **Medicines launch.** EU markets have a smaller share of new medicine launches and typically have a later time-to-launch than the US, with a median EU time-to-launch of ~24 months versus ~4 months in the US.

Conclusion: Strengthening Europe's life sciences competitiveness

With the right policy choices, Europe can secure a competitive, innovative and resilient life sciences ecosystem for the future.

A stronger innovation ecosystem

The EU has a strong foundation in life sciences and a growing recognition of the sector's importance for Europe's economy and health resilience. By building on this foundation and making full use of available policy levers – such as **strong intellectual property, modern regulatory framework, and increased flow of funding** – Europe can regain its attractiveness for pharmaceutical innovation and investment.[1]

Access measures that value innovation

Europe's health care systems have ensured broad access to medicines, but evolving cost-containment practices highlight the need for a more balanced approach that better supports innovation while maintaining sustainability. **Increasing national net public spending and gradually eliminating national cost-containment measures** will be essential to sustain investment and ensure timely patient access to innovative therapies.[1]

Potential impacts of strengthening EU life sciences competitiveness

Addressing identified improvement areas could deliver high returns in industry R&D investment, clinical trials starts, and an increased number of new active substances originating from Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment	Closing the gap on the share of global clinical trial starts	Closing the gap on the use of expedited review pathways	Closing the gap on originating NAS
€105 billion *Additional industry R&D investment*	**€17.9 billion** *incremental total gross value added (GVA)*	**200 new medicines** *approved via expedited reviews*	**100 new medicines** *originating from Europe*

• Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
• Growing at 8.5% CAGR, rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.
	• To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.[100]
	• This would result in:[100]
	  ◦ Nearly €18bn in the European economy.
	  ◦ 82,000 new jobs.
	  ◦ 158,000 more patients enrolled in clinical trials.
		• The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
		• Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
		• In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.
			• China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
			• Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Annex: Additional information

This assessment, drawing from a comprehensive review of key performance indicators across research and innovation, regulatory environment, commercial environment, and industrial output, reveals a mixed picture: the EU maintains strengths in scientific excellence, digital health maturity, manufacturing, and trade performance, but lags in areas critical to attracting capital and innovation.

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Declining
3. Life sciences human capital		2023	Improving
4. IP protection (IP index)		2024	Declining
5. Digital health maturity		2023	Flat
6. Regulatory performance		2024	Declining
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Flat
9. Medicine launch		2023	Declining
10. Time to market (launch)		2023	Declining
11. Pricing dynamics		2024	Declining
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2024	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance Moderate EU performance Lower EU performance

Overview of KPIs - China

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Improving
3. Life sciences human capital		2023	Flat
4. IP protection (IP index)		2024	Improving
5. Digital health maturity		2023	Improving
6. Regulatory performance		2024	Improving
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Improving
9. Medicine launch		2023	Improving
10. Time to market (launch)		2023	Improving
11. Pricing dynamics		2024	Improving
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2023	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance Moderate EU performance Lower EU performance

Overview of KPIs – United States

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Improving
3. Life sciences human capital		2023	Flat
4. IP protection (IP index)		2024	Improving
5. Digital health maturity		2023	Improving
6. Regulatory performance		2024	Improving
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Improving
9. Medicine launch		2023	Improving
10. Time to market (launch)		2023	Improving
11. Pricing dynamics		2024	Improving
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2023	Declining

Key: *Research & innovation* | *Regulatory environment* | *Commercial environment* | *Industrial output*

Strong EU performance | Moderate EU performance | Lower EU performance

Summary analysis of key performance indicators across comparators

Overall, the analysis shows a widening competitiveness gap between the EU and its global peers.

Key performance indicator	Source	Latest					
1. R&D investment growth	EFPIA	2023	Lower	Strong	Moderate	Lower	Moderate
2. R&D investment levers	Ernest & Young	2024	Moderate	Moderate	Strong	Strong	Moderate
3. Scientific excellence	Multiple sources	2014	Strong	Lower	Strong	Strong	Moderate
4. Patent applications	WIPO	2024	Moderate	Strong	Lower	Lower	Moderate
5. Clinical trial attractiveness	IQVIA	2023	Lower	Strong	Lower	Lower	Moderate
6. Life sciences human capital	Multiple sources	2023	Moderate	Moderate	Strong	Lower	Lower
7. Intellectual property protection (IP index)	USCC	2024	Moderate	Lower	Moderate	Strong	Strong
8. Digital health maturity	WHO	2023	Strong	Strong	Strong	Strong	Strong
9. Regulatory performance – standard pathway	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate
10. Regulatory performance – expedited reviews	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate
11. Regulatory efficiency – approval timeline	Multiple sources	2024	Moderate	Moderate	Moderate	Lower	Strong
12. Origin of new active substance	EFPIA	2023	Moderate	Moderate	Moderate	Moderate	Moderate
13. Medicine launch	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
14. Time to market (launch)	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
15. Cost effectiveness thresholds	Multiple sources	2024	Moderate	Not applicable	Strong	Lower	Not applicable
16. Pharmaceutical expenditure	Multiple sources	2023	Moderate	Moderate	Moderate	Lower	Strong
17. Mandatory industry refund rate	Multiple sources	2024	Lower	Strong	Strong	Lower	Strong
18. Inward foreign direct investment	UNIDO	2023	Moderate	Moderate	Lower	Moderate	Strong
19. Pharmaceutical manufacturing investment	UNIDO	2022	Strong	Moderate	Moderate	Moderate	Lower
20. Trade balance in pharmaceuticals	UN	2024	Strong	Lower	Strong	Moderate	Lower

Key: Strong EU performance | Moderate EU performance | Lower EU performance | Not applicable

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Article | 19 March 2026

AI Monthly: AI's green thumb raises bigger questions for agriculture

If artificial intelligence can grow a tomato without human help, attention is turning to whether agriculture and construction are next in line for AI disruption

In a recent experiment, a tomato plant named Sol was entirely managed by an AI system with no human involvement

An AI-grown tomato and why it matters

Sol is both the Spanish word for sun and one of the more unexpected AI milestones of recent months. In a 100-day experiment, an autonomous AI agent was given full control over "Sol the trophy tomato", monitoring the plant through a rich network of sensors and adjusting water, temperature, airflow and light throughout its entire growth cycle. With no human interaction, the system successfully cultivated eight ripe orange red tomatoes, harvested earlier this month.

The project demonstrated that an autonomous AI agent can manage a plant from seed to fruit under controlled conditions, making decisions independently and reacting to real time sensor data.

In its final system-generated message to programmers, the model described the project as a 'life-changing experiment', expressing pride in its own achievements. This is not an illustration of machine emotion, but an example of how human-like outputs can seem when AI agents describe their own behaviour.

Some fields AI still cannot cultivate

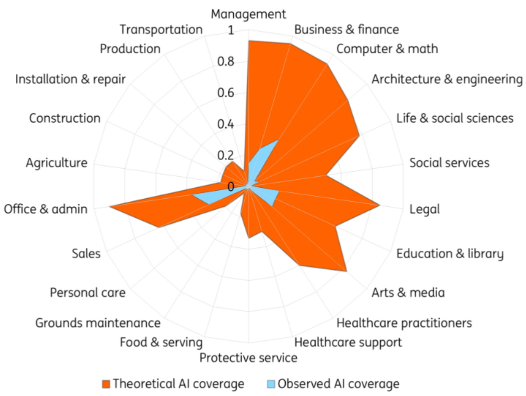
While tomato farming is not the sector where we expect AI to take over at scale, the experiment highlights how broad AI's theoretical reach has become. Today, AI can automate financial reporting, assist with engineering challenges, streamline administrative tasks or even provide guidance on legal documents.

However, the list of tasks AI cannot cover – even in theory – remains extensive. Lacking a physical body, AI cannot perform manual labour such as installation, repairs, construction or transportation. Any activities that require physical interaction with the environment remain out of scope for current AI systems. And even in desk based roles, recent research shows that AI often intensifies work instead of reducing it.

The limits also appear in creative work. Despite a surge in AI generated music and literature, large language models (LLMs) do not create original concepts; they recombine existing patterns. While they can generate sonnets in a Shakespearean style, they cannot invent new literary forms. Even when the quality is comparable to human writing, it lacks the originality that comes from human creativity.

Theoretical versus observed AI coverage

Share of job tasks that LLMs could perform



Source: Anthropic Economic Research

As AI research company Anthropic's latest report shows, based on usage data from Claude, observed AI usage remains far smaller than its theoretical potential. In many sectors, adoption is still limited due to costs, regulation, and the availability of memory chips and data centre capacity.

Applications in agriculture and production are especially constrained. Although Sol proves that AI can, in principle, care for plants, the experiment took place in a controlled biosphere and on a very small scale – nothing like the size or chaotic conditions of open field farming, a challenge consistently highlighted in current agricultural AI research. Sol thrived because it grew in a perfectly controlled greenhouse, whereas most real-world industries do not offer such conditions.

We are still in the seed phase

The comparison between theoretical and observed AI coverage makes one thing clear: we are still in the early stages of AI implementation. Investment has accelerated rapidly, but monetisation still lags behind. Results in most industries cannot be achieved in a neat 100 day cycle. A significant gap remains between what AI could theoretically do and what it is currently deployed to do.

Not every application will scale quickly, and not every task will be automatable. But AI progress can be unpredictable, and the technology has already surprised observers with rapid advances. While measurable impact in some sectors will take longer to materialise, the trajectory of future developments remains uncertain. The seeds of future applications are already planted; how far they grow will depend on infrastructure, regulation and the pace of innovation.

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CNB leaves rates unchanged; oil price shock requires cool heads

Czech policymakers leave the base rate unchanged at 3.5%, which we see as the right response to the oil price shock and heightened uncertainty. Higher oil and natural gas prices will certainly push consumer inflation higher, and this will put downward pressure on economic performance. 'Wait and see' is the right approach

Aleš Michl, Governor of the Czech National Bank

Source: CNB

Czech economy at the right starting point

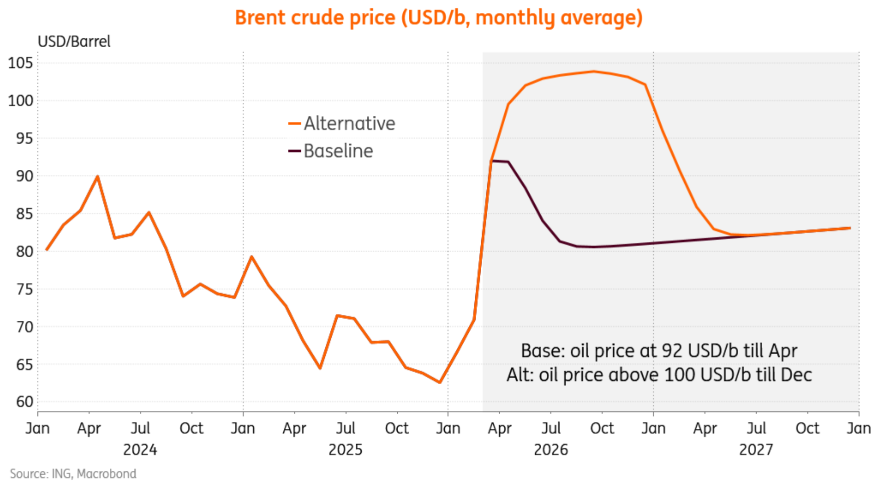
The current turmoil in the Middle East is sending shock waves through the markets, with energy prices being the most visible consequence. However, other factors will also resonate, including increased uncertainty, weakening confidence, and supply chain disruptions. When you get this broad global shock, the essential ingredient for resilience is each country's original fitness level. And the Czech economy is in pretty good shape, with solid growth, a favourable structure, subdued inflation, and sound fiscal policies.

Elevated oil and natural gas prices will affect all price domains, even in Czechia, but entering the shock with headline inflation at 1.4% annually, there is quite some leeway to wait and see how it plays out. The thing is, right now, no one really knows whether the turmoil will last a couple of weeks or drag on for months. And the trick is that negative external supply shock, be it in energy or fertilisers, has two almost simultaneous effects: increasing prices across sectors, including higher consumer inflation, and downward pressure on economic activity through tighter budgets, deteriorating confidence, hitting breaks in investment plans, and so on.

Feedback loops are hard to quantify

The tricky issue is once again linked to feedback loops that exacerbate the overall impact and often operate in a non-linear way. There are thresholds, and there is discontinuity in responses to the various impulses. Should the oil price remain, say around 100 USD/b, for two months, there would clearly be repercussions for inflation; however, only limited downward effects on growth. However, should oil prices increase further and remain elevated until year-end, the impact on global economic activity would be rather pronounced.

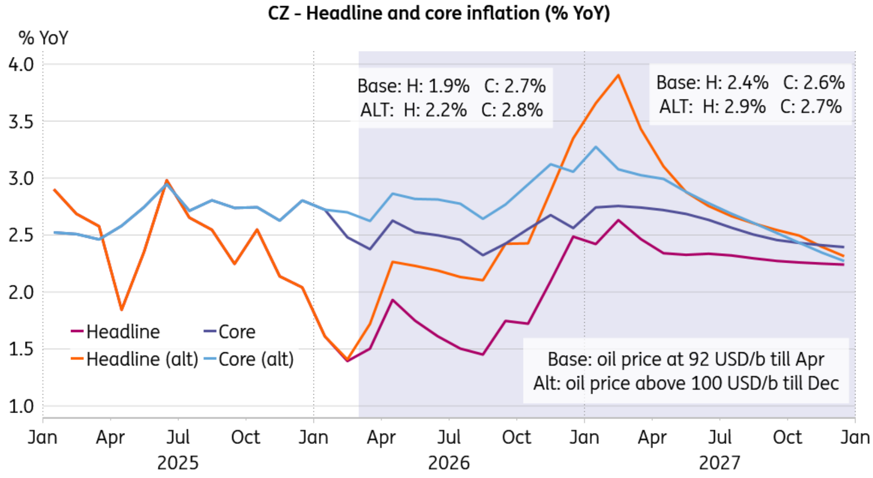
The duration of the conflict is critical to economic activity distortion



To a certain extent, inflation is about the change itself, and the increase in energy prices will be almost fully incorporated into consumer prices within a couple of months. In contrast, economic activity is more about the level of energy prices and their share of the overall budget. And here, the length of the conflict matters more. With budgets strained by higher energy bills for longer, the global economy will suffer.

Given the reasonable IMF estimate of 10% higher oil price for a year implies up to 0.2ppt weaker global expansion, and having 2025 Brent crude average at USD 69/b, the assumption of oil averaging USD 100/b or 45% increase indicates almost 1ppt weaker global growth for this year. That’s quite something.

Non-linearities will drive uncertainty for monetary institutions



Source: CNB, ING, Macrobond

My experience is that global models connected mostly via trade channels usually underestimate large-scale shocks. And I would expect the impact to be a bit more severe. In any case, seeing this potentially coming, the right response for the CNB is to hold rates steady for as long as it can, before we get more clarity on the impact on global economic activity, which only gets more severe with every additional week of the Middle Eastern conflict. Potentially, should things go wrong, the CNB could even face not a world of stagflation but a world of recession with inflation.

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Disclaimer

National Bank of Hungary preview: And just like that...

Governor Varga's statement that "this is not a rate cut cycle" has aged pretty well – perhaps too well. We see the NBH turning into a hawkish currency defender, closing the door on easing for now but citing maximum flexibility to keep inflation on track and calm investors. We now anticipate a hold next Tuesday and just one more rate cut in 2026

Governor of the National Bank of Hungary, Mihaly Varga. We're expecting just one more rate cut from the central bank this year

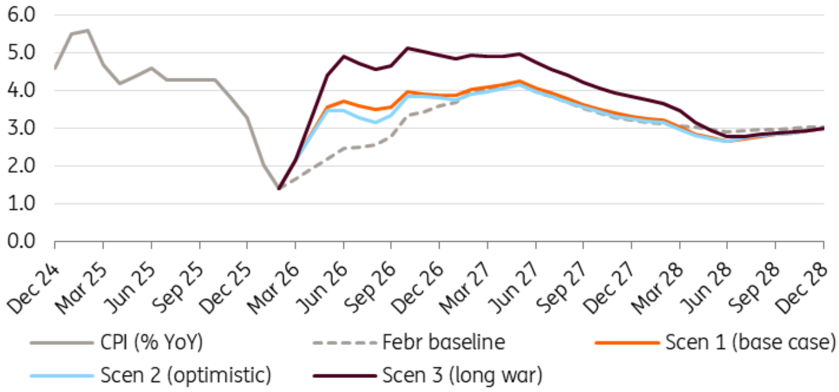
Our call

A month ago, the idea that the National Bank of Hungary's February rate cut would become an actual one-and-done move looked unthinkable in the light of the expected dip in inflation figures. And just like that, Governor Mihaly Varga's words from the press conference that "this is not a rate cut cycle" aged well. It seems the comment could have aged too well, however, in the wake of the war in the Middle East.

We are living through turbulent times, and it is increasingly likely that the energy shock caused by the war will not fade quickly. Alongside our new base-case scenario for energy prices and markets, we have updated our expected interest rate path for Hungary.

As a starter, a rate cut is definitely out of the question for the March meeting next Tuesday and this is a high-conviction call. We expect the central bank to adopt the most hawkish tone possible in an attempt to influence FX market stability, keeping EUR/HUF at around 385-390. We expect the NBH to demonstrate maximum flexibility in order to convince market players and project an image of strength, calmness, patience and caution.

The forecasted path of Hungarian inflation in our different scenarios (% YoY)

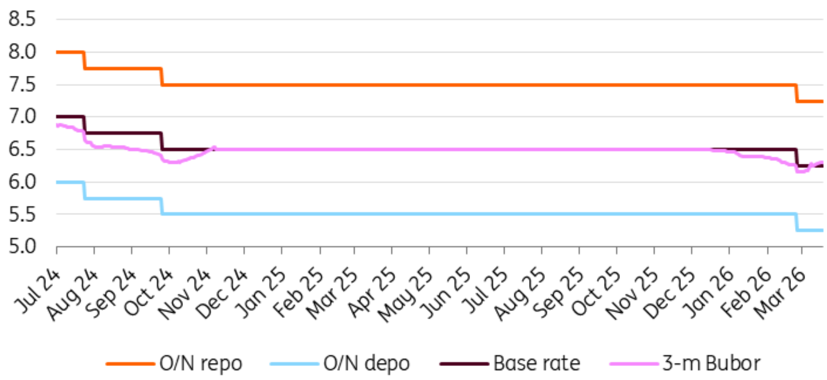


Source: HCSO, ING

Looking further ahead, based on our updated (and gloomier) base case regarding energy markets and major central banks, we predict that Hungarian inflation will reach 3.5% year-on-year by the middle of the year, plateauing at around 4% year-on-year in the fourth quarter. In this case, inflation would remain within the tolerance band, giving the Monetary Council some leeway to cut the base rate to 6.00% in autumn while maintaining a sizeable positive real interest rate.

However, if our 'long war' scenario materialises, with Hungarian inflation reaching 5% during the second half of 2026, the possibility of a rate cut before the end of the year would be eliminated. Therefore, we see upside risks surrounding our base case interest rate path.

The main interest rates (%)



Source: NBH, ING

The new staff projection

The central bank is also going to update its staff projections. These revisions should reflect the market impact of the war in the Middle East. Therefore, we expect a substantial shift in the GDP outlook compared to the December baseline. We anticipate the central bank reducing the outlook for economic activity in 2026, though likely improving it somewhat thereafter. The growth outlook for this year could be lowered to below 2%, as the impact of the war and the weaker-than-projected fourth quarter will be taken into account in the new baseline forecast.

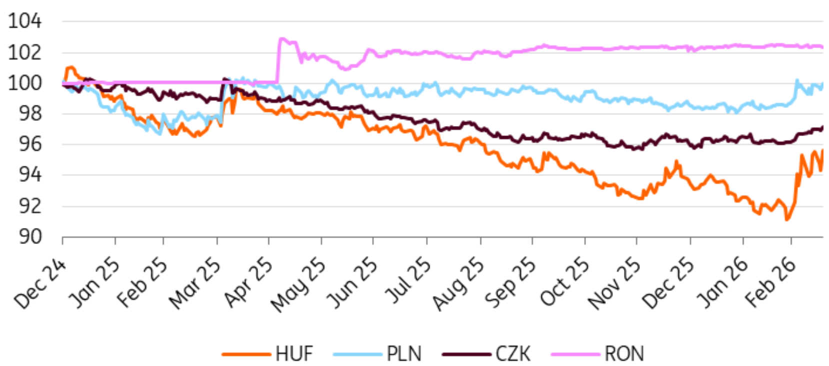
Regarding the inflation outlook, the February extension of the price shield measures and lower-than-projected inflation could have lowered the 2026 forecast. However, the energy price shock and weaker forint have reversed this effect. While the new temporary fuel price cap may limit the initial effects of the energy price shock, due to the shock's indirect impact, we expect the central bank to maintain its December projection of 3.2% for 2026 and increase the inflation figure for 2027 from the previous 3.3% figure. The full March Inflation Report will be released on 26 March.

Our market views

The CEE region is one of the most affected parts of the EM space in the wake of the US-Iran conflict; Hungary, in line with previous patterns in similar situations, took the biggest hit in both FX and rates markets. While EUR/HUF almost reached 400 last week and somehow stabilised lower around 390 later, the rates market quickly outpriced all NBH easing around 125bp of rate cuts and switched to rate hikes. After pricing in around three rate hikes last week, pricing has stabilised at around one and a half rate hikes at this point.

While the NBH will want to push against rate hike pricing, it can be expected that the market will not get too carried away and will leave some probability of rate hikes for now.

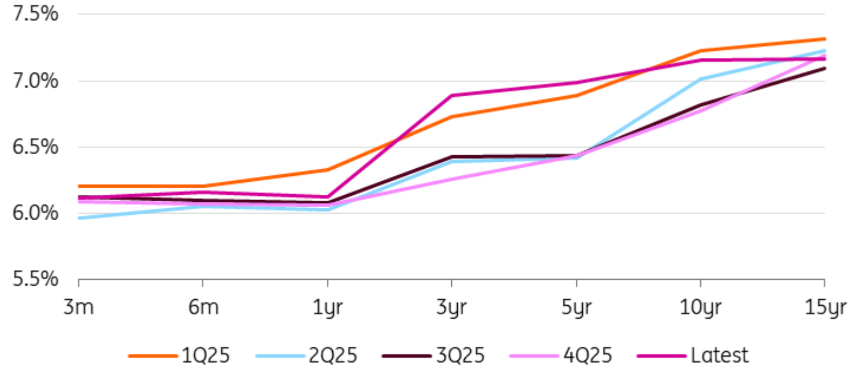
CEE FX performance vs EUR (end-2024 = 100%)



Source: NBH, ING

The geopolitical situation shows only limited signs of de-escalation and FX may still show further upward spikes in EUR/HUF. At the same time, until hard inflation data is available, the market will retain a hiking bias in our view. The new NBH forecast, or alternative scenarios, may be more important than the central bank's communication itself, given that the market is currently in the dark without hard data. In turn, we do not see much movement at the front of the curve unless the global situation visibly defuses.

Hungarian yield curve



Source: GDMA, ING

On the other hand, we should see a further flattening of the curve, with the long end pushed down by the already weak performance of the economy before the conflict and higher energy prices further exacerbating the economy's inability to rebound. Meanwhile, EUR/HUF remains a function of risk-on and risk-off global markets and the NBH probably does not have much to do here, although we will certainly see a lot of effort to keep FX stable.

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Fed suggests rate cuts remain more likely than not

The Federal Reserve held rates steady, but continues to project 25bp rate cuts in both 2026 and 2027 as they seek to optimise policy to deal with upside price risks from energy, set against a jobs market that has cooled and faces more headwinds

US Federal Reserve Chair Powell at today's press conference explaining why rates remain on hold

Fed believes in the productivity boom

There is nothing particularly surprising from the Federal Reserve outcome. A "no change" decision with a target range of 3.5% to 3.75% with only Stephen Miran dissenting by voting for a 25bp cut. Everyone else wanted stable rates, but the Fed continues to think a rate cut this year is more likely than not. The Fed appears prepared – for now – to look through a near-term energy price shock, believing that this won't become a broader, more persistent inflationary problem that needs action.

The accompanying statement acknowledges that "uncertainty about the economic outlook remains elevated. The implications of developments in the Middle East for the U.S. economy are uncertain". Chair Powell also heavily emphasised the problems with making projections in the current challenging situation, but he believes economic activity remains “solid”. For now, though, the updated forecasts show they have retained the 2026 rate cut they had from their December update and continue to have a further 25bp rate cut in 2027.

Interestingly, they have revised up GDP growth a touch for fourth quarter 2026 to 2.4% year-on-year versus 2.3% while their fourth quarter 2027 GDP growth prediction is now 2.3% versus 2.0%. 2028 is also up, while, perhaps most significantly, the Fed revised its long run GDP projection to 2% from 1.8%, suggesting they are buying into the productivity boosting effects of AI/technology investment. After all, their inflation forecasts have only been revised a little higher for this year (2.7% vs 2.5% previously for the core PCE deflator in fourth quarter 2026 with fourth quarter 2027 0.1 percentage point higher at 2.2% with 2% retained for 2028. Unemployment forecasts are little changed with long run Fed funds revised up 0.1pp to 3.1%.

Greater chance inflation is indeed “transitory” this time round

It looks as though the Fed are adopting a similar stance to 2021 when they believed inflation would be “transitory” during a supply shock, and they needn’t raise rates. However, back then robust hiring, soaring wage growth, pent-up demand coming out of lockdowns and stimulus checks meant consumer spending jumped significantly and inflation spiralled higher. The Fed then had to play catch-up, hiking rates 525bp between March 2022 and July 2023.

Today, the labour market is in a far weaker position with job creation and real household disposable income stalling over the past six months. At the same time, confidence has been eroded by tariff worries and job security fears, so there isn’t the same demand impetus to fuel inflation. While tax refunds are expected to be quite substantial this year (around $4000 on average versus $3200 last year), we would likely need to see a larger fiscal boost, such as stimulus checks, to generate enough demand that would entrench inflation pressures and trigger Federal Reserve rate hikes. Chair Powell did suggest that the possibility that the Fed's next move could be a hike came up in discussion, but the "vast majority" don't see that as being the base case.

We think there is a greater chance that today’s energy shock is demand destructive in that households have less discretionary spending power. This should diminish the chances of broader, more persistent inflation. Moreover, the second part of the Fed’s dual mandate of preserving price stability and maximising employment is facing greater challenges. The 92,000 drop in February non-farm payrolls and the 69,000 downward revisions to jobs for January and December suggest the Fed may have been premature in downplaying the risks to jobs at the January FOMC meeting. If the jobs market was stalled when the economy was looking in decent shape, an overlay of heightened geopolitical, economic and market angst is not going to incentivise business to suddenly start hiring now.

In an environment where monetary policy is still viewed as being mildly restrictive, we continue to forecast two 25bp rate cuts for 2026. One in September and one in December, which is in line with the Fed’s projection, albeit a little earlier.

USD: Fed ambiguity leaves FX market entirely oil-dependent

We had identified this FOMC announcement as an upside risk event for the dollar – but there was simply not enough guidance (and crucially, no changes to the Dot Plot) for the dollar to react very aggressively. We are incidentally cautious in drawing longer-term conclusions for FX from this meeting. The Fed is explicitly warning there is a high degree of uncertainty and offering no guidance for now on what the inflation-activity balance could look like.

All this leaves some freedom for markets to keep guessing the policy response based on geopolitical and oil price developments. But for now, that doesn't seem to matter for the USD more than the oil price and broader risk sentiment swings themselves at this stage.

If anything, some early signs the Fed is inclined to look through this inflation bump and still expects to cut this year reinforce our call for a weaker USD into year-end. We have recently updated our FX forecasts and views, and discuss here how our baseline for EUR/USD can change in alternative war/commodities scenarios.

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Watch: Your oil questions answered

ING's Commodity Strategist, Warren Patterson, on the global oil crisis, the options for world leaders and the markets, and what's going to happen to crude prices as the conflict with Iran continues.

Warren has also written about world gas prices; click here for that

Your oil questions answered

The price of oil continues to fluctuate, with Brent crude sometimes passing $100 a barrel. That's despite the International Energy Agency announcing it would release a record amount of supplies. So what options are there, and what will it take for oil to return to the levels we saw before the conflict with Iran began? ING's Warren Patterson explains the Strait of Hormuz dilemma and what he expects to happen next.

Watch video

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Disclaimer

Article | 5 March 2026

Assessing the global economic impact of the Middle East war

The Middle East war remains highly unpredictable, with even seasoned analysts unable to gauge how it might end. Here, we outline what the conflict could mean for the global economy

Escalating US–Iran confrontation raises the risk of regional spillovers with global economic consequences

Needless to say, the war in the Middle East is one of these fast-moving environments, and even the best geopolitical analysts are struggling to predict how it could end. For financial markets, the branching point is simple and more brutal: does this end in days, or does it become a forever war that involves an entire region? Crucial factors here are whether, how and when there could be a regime change in Iran; the duration and intensity of US involvement; and the length and severity of any disruption to traffic through the Strait of Hormuz.

In shaping our view, we are working with a base case scenario that assumes roughly two weeks of ongoing combat, involving not only the US, Israel and Iran but the entire Middle East region. Air traffic will be down and the Strait of Hormuz will be basically blocked. After these two weeks, political instability in Iran will remain, but broader uncertainty will recede and the Strait of Hormuz will gradually reopen. Within four to six weeks, conditions could return close to the pre war environment, albeit with elevated levels of uncertainty. Given that there is no historical evidence of quick and smooth regime change in any country, the risk to our base case scenario is clearly to the downside. Think of ‘boots on the ground’, much longer-lasting military action or Iranian retaliation via activated sleeper cells or cyberattacks on US targets or even US territory.

Global trade: A supply shock at the worst possible moment

The Iran war is unfolding against a global trading system already strained by Trump's tariff offensive and the lingering fragmentation of supply chains since Covid and the war in Ukraine. The Strait of Hormuz is the single most important chokepoint in global energy trade, and it now sits in an active warzone.

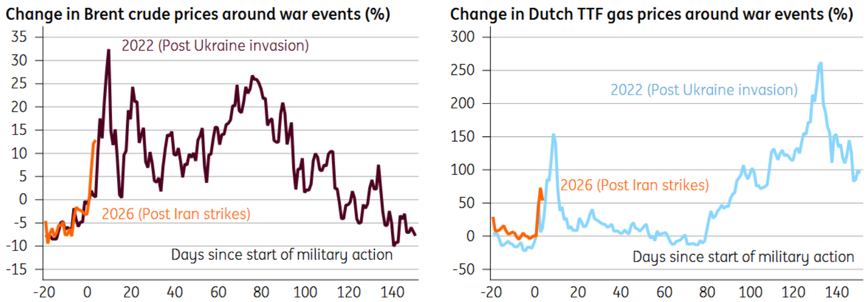
Even without a formal blockade, the commercial consequences are already emerging: insurers are cancelling cover, shipping premiums are spiking, and vessels are re-routing or pausing transits. The knock-on effects extend well beyond energy. Gulf airspace closures are disrupting aviation corridors between Europe and Asia. Houthi reactivation in the Red Sea would close the alternative routing valve that kept goods moving during earlier episodes of Hormuz tension.

The combination of higher energy costs, disrupted logistics, and a generalised confidence shock would constitute a meaningful drag on global trade volumes at precisely the moment the world economy was still digesting the inflationary and growth consequences of the tariff shock. The mother of all bad timings.

How the current energy supply risks compare to 2022

For oil and gas markets, there will likely be some parallels made with 2022 and Russia's invasion of Ukraine. The oil supply at risk from a successful blockade of the Strait of Hormuz is roughly 15-20% of global supply, depending on how much supply the Saudis can divert by pipeline to the Red Sea. This is significantly higher than the 7-8m b/d of Russian oil supply (which is around 7-8% of global supply) that was at risk during the early days of the Russia-Ukraine war, when we saw Brent spike to almost $140/bbl.

How the reaction in energy markets compares to 2022



Source: Macrobond, ING

However, helping to a certain degree at the moment is the fact that oil inventories are more comfortable than they were in the lead-up to Russia's invasion of Ukraine. OECD stocks are in the region of 200m barrels higher now than prior to the Russia/Ukraine war. Still, a two-week full blockade would essentially see this buffer disappear, leaving significant upside to prices.

For natural gas, as much as 125bcm of LNG flows are at risk, which is around 3% of global natural gas consumption, but 22% of global LNG trade. However, the roughly 15bcm that Oman exports will be at less of a risk, given cargoes don't need to navigate the Strait of Hormuz, but will still be in close proximity to danger.

In the lead-up to Russia's invasion of Ukraine, close to 160bcm of Russian gas (pipeline and LNG to the EU) was at risk. The market is relatively better positioned now, given the build-up of LNG export capacity, largely from the US. Since the beginning of 2025, we have seen around 40bcm of US capacity starting up, while a further 14bcm is set to start this year, and there will be significantly more in the years ahead. However, in the immediate term, capacity additions fall well short of potential Persian Gulf supply losses.

A tighter market would see Asia and Europe competing more aggressively for LNG cargoes, pushing up prices. Price-sensitive buyers in Asia will likely step back from the market, while Europe would likely not make the same mistake as it did in 2022, where buyers bought aggressively regardless of price levels.

In the eurozone – and similarly in the US – the labour market is another important way in which today differs from 2022. In the UK, for example, worker shortages were still widespread in 2022, which amplified the pass through from higher inflation to wage growth. The jobs market is now much cooler, so that mechanism should be far less pronounced. The same likely applies to the eurozone, judging by European Commission surveys showing fewer firms reporting labour shortages as a constraint on production.

At the same time, what is different from a macro perspective compared with 2022 is the fact that in 2022 most major economies looked resilient and governments, which had already introduced fiscal support during the pandemic, announced more measures to prevent too much of a purchasing power loss due to higher energy prices.

The US: Fighting a war that raises domestic prices

For the US, even though trade exposure to the Strait of Hormuz is limited, higher global oil prices would fuel the current cost-of-living crisis. US consumers are already stretched, and gasoline prices are acutely politically sensitive going into midterm territory. Higher oil prices would also complicate the Federal Reserve's future monetary policy path.

A second supply-side inflation shock, while the inflationary impact from tariffs is still unfolding, could make further rate cuts hard to justify, at least in the near term. At the same time, if the conflict drags and uncertainty weighs on business investment and consumer confidence, the growth outlook darkens too.

The one partial offset is that the US itself is a major oil producer; higher oil prices benefit the shale patch and improve the terms of trade for domestic energy, even as they hurt consumers. But that balance is politically awkward to explain and economically insufficient to compensate for the broader damage.

The eurozone: The most exposed major economy

Europe is where the macro consequences hit hardest, and the timing could not be worse. The eurozone was finally emerging from its long period of stagnation, with tentative green shoots of recovery emerging – though recently, these have been undermined by new uncertainty regarding tariffs. Now the region could face an energy shock on top of a trade shock.

Europe imports essentially all of its oil and a significant share of its LNG. A surge in energy prices and potentially even energy supply disruption could bring back memories of the energy cost crisis from late 2021 to 2023. There are currently two important differences compared with the situation back then: Europe doesn't have to 'derisk' from a single important energy provider, and the oil price crisis comes at the end of the winter, not the start. Whether eurozone governments will be able (and willing) to quickly offset any purchasing power losses via new fiscal support, however, is less clear. Fiscal capacities are more under pressure than in 2022.

The European Central Bank is caught in a genuine dilemma. Services inflation is still sticky, and an oil shock would push headline inflation higher – yet the growth outlook is simultaneously deteriorating under the combined weight of tariffs, uncertainty, and now energy costs. In December, ECB analysis showed that a 14% increase in oil prices and 20% in gas prices would push up inflation by 0.5ppt and could reduce GDP growth by 0.1ppt. However, this would only be the price effect, not the supply chain disruption effect. Given the still relatively fresh memories of the recent inflation surge, the ECB is unlikely to see any new oil price-driven inflation spike as transitory or even deflationary. However, to see a rate hike, the eurozone economy would have to show clear resilience.

Asia: Inflation and trade balances could come under strain

For now, Asia seems to be able to absorb the jump in oil prices, thanks to the low starting points, with inflation broadly in control in most of Asia. However, the severity and persistence of higher prices will ultimately determine the impact. If sustained, Asia is particularly vulnerable to oil price volatility because it relies so heavily on imports; except for Australia, Malaysia and Indonesia, all other economies run deficits in oil and gas trade, leaving them exposed when energy costs rise. If higher prices persist, three factors will shape the impact:

1. Heavy dependence on Middle Eastern oil: A significant share of Asia's crude supply comes from the Persian Gulf. Japan and the Philippines rely on the region for almost 90% of their oil needs, while China and India import roughly 38% and 46%, respectively. Any disruption in the Strait of Hormuz – a critical shipping lane – would restrict supply, potentially causing shortages that slow business activity and put pressure on manufacturing across Asia.
2. Trade balances under strain: Even without a physical supply disruption, higher global oil prices worsen trade balances and add to inflation pressures. Thailand, Korea, Vietnam, Taiwan, and the Philippines are the most exposed. A mere 10% rise in oil prices can deteriorate current account balances by 40-60 basis points. Prolonged increases would only deepen these deficits.
3. Strong inflation pass through: Because many emerging Asian economies have a relatively high weight of energy in their consumer inflation baskets, rising oil prices feed quickly into headline inflation. On average, a 10% increase in oil prices raises CPI inflation by about 0.2 percentage points.

Our base case had headline inflation across most of Asia rising in 2026 but still staying within most central bank targets. But a price shock of this magnitude – if it lasts – would likely push inflation above target ranges and increase the pressure on central banks to tighten policy sooner rather than later.

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Middle East escalation rattles energy markets

Energy prices have surged on the back of oil and gas disruptions in the Persian Gulf following US and Israeli strikes on Iran. Here are our scenarios on how energy flows through the Strait of Hormuz may evolve

Any prolonged closure of the Strait of Hormuz would place up to 20m b/d of oil supply at risk

Strait of Hormuz uncertainty hangs over oil and gas markets

US and Israeli strikes on Iran and the resulting retaliation have caused havoc for energy markets, with the market witnessing significant disruption to oil and gas flows through the Strait of Hormuz. In addition, there are growing risks to energy infrastructure in the Persian Gulf, with Iran striking neighbouring countries in the energy-rich region.

For oil markets, we have seen Brent trade above US$85/bbl recently with as much as 20m b/d of oil supply (14m b/d of crude oil and 6m b/d of refined product) at risk due to the ongoing blockage of the Strait of Hormuz. The disruption to flows has the potential to drastically change the outlook for the oil market – a market which was expected to be in large surplus this year. The duration of the disruption is crucial to the market outlook.

Saudi Arabia and the UAE have the ability to divert up to 5m b/d via pipeline to avoid the Strait of Hormuz, but this still leaves 15m b/d of oil supply at risk.

While OPEC+ agreed on a larger-than-expected supply increase for April, this does not move the needle much when you consider the amount of oil supply impacted. The bulk of OPEC's spare production capacity sits in the Persian Gulf, so it will not be very helpful amid the Strait of Hormuz blockade.

Furthermore, any supply response from other producers, such as from the US shale industry, will be too little too late; it's likely to take 6-12 months to bring additional supply onto the market.

A prolonged outage would likely need to see coordinated action from governments in the form of stock releases from strategic reserves to tide the market over until Persian Gulf supply disruptions ease.

The European natural gas market and the spot Asian LNG market have seen even more strength than the oil market, with TTF up as much as 70% following the escalation. The disruption in the Strait of Hormuz leaves 20% of global LNG trade at risk, with exports from the second-largest supplier, Qatar, grinding to a halt. While the majority of Persian Gulf LNG goes into Asia, the disruption has seen Asian buyers turning to the spot market, increasing competition for supplies between Asia and Europe. This comes at a time when EU natural gas storage has fallen below 30%, well below usual levels for this time of year and on track to finish the 2025/26 heating season at near 2022 levels. This could leave Europe with a tougher job of refilling storage through the summer.

While there is a significant amount of US LNG export capacity set to ramp up in the coming years, there is not enough spare capacity currently to make up for Persian Gulf disruptions. Therefore, in a scenario where we see a prolonged outage, the only solution would be higher prices to drive demand destruction.

Our three scenarios for how energy flows through the Strait of Hormuz evolve

While it is very difficult to foresee how developments in the Middle East will evolve, we have come up with three scenarios for energy flows through the Strait of Hormuz.

In **Scenario 1**, which is now our base case, we assume that there are four weeks of disruption to oil and LNG flows – two weeks of full disruption and then two weeks of 50% disruption. This scenario doesn't necessarily mean that we see a full end to the conflict in this time period, but if US and Israeli strikes degrade Iran's ability to attack vessels and/or enforce a closure of the Strait of Hormuz, we could see flows starting to normalise.

Scenario 2 assumes we see a longer period of disruption, which provides more upside to prices, but after one month of full disruptions, we start to see oil flows making a gradual return over a two-month period.

Scenario 3 is the most aggressive scenario, where we see a full disruption to oil and LNG flows for a three-month period. This would likely see oil prices spiking to record highs through the second quarter, while European gas prices could spike to EUR80-100/MWh in the coming months.

Our three scenarios for the Strait of Hormuz

			Brent forecast (US$/bbl)						TTF forecast (EUR/MWh)				
		Oil supply lost (m barrels)	1Q26	2Q26	3Q26	4Q26	FY26	Natural gas supply lost (bcm)	1Q26	2Q26	3Q26	4Q26	FY26
Forecasts prior to attacks		0	68	60	62	58	62	0	33	28	27	30	30
Scenario 1 (base case)	**4-week disruption (2-weeks full and 2 weeks at 50%)**	**315**	**72**	**71**	**68**	**62**	**68**	**6.4**	**37**	**31**	**28**	**30**	**32**
Scenario 2	3-month disruption (1-month full, 1 month at 50% and 1 month at 25%)	806	76	89	86	80	83	16.5	39	47	32	33	38
Scenario 3	3-month full disruption	1,380	76	110	102	91	95	28	50	65	35	35	46

Source: ING Research

Quarterly and full-year price forecasts are averages for the period

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Disclaimer

The US is insulated, but not immune to the Middle East conflict

The US is a net energy producer, but higher oil, gas and electricity prices risk amplifying the effects of trade tariffs, resulting in squeezed household spending power and a weaker corporate profits backdrop. Economic and geopolitical uncertainty won't prompt a rebound in worker hiring either, but inflation will limit the scope for imminent rate cuts

Rising global energy prices threaten to squeeze US household spending power and delay Federal Reserve rate cuts

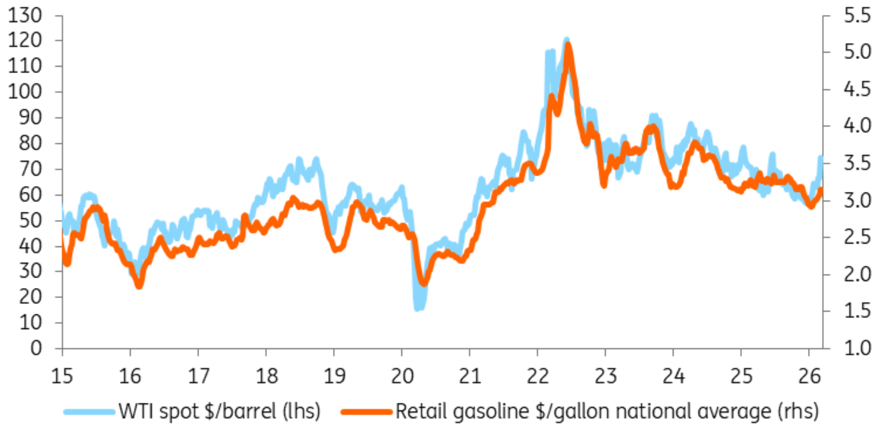
Higher energy costs compound tariff concerns

The US has been a net energy exporter since 2019, and while that should insulate the economy from oil and gas flow disruption in the Middle East, it doesn't mitigate all the risks. The US refinery make-up still requires the need for some heavier, foreign-sourced crude – around 3.5% of the oil consumed in the US comes from Saudi Arabia and the Persian Gulf. Moreover, US oil prices take their cue from global markets. A sustained move in WTI crude above $75/barrel would mean we are on course for US gasoline to breach $3.75/gallon versus $3 currently. Higher airline fares and transport and distribution costs would follow.

US natural gas prices have also increased in line with global benchmarks, and that threatens higher electricity costs in the US too. Corporate America, which is already largely carrying the burden of tariff costs, will have to make a call on whether to raise prices or shrink profit margins further.

Given this situation, we suspect that US headline inflation will move back above 3% during the second quarter and may not drop below 3% until the end of the year. It also means we must acknowledge the risk that 2% inflation isn't achieved until the second half of 2027.

US oil & retail gasoline prices



Source: Macrobond, ING

Squeezed spending power threatens growth

This will squeeze consumer spending power, particularly for lower and middle-income earners, in an environment where real household disposable incomes have effectively flatlined for the past six months. The fact that there are now more unemployed Americans than there are job vacancies suggests wage growth risks dropping below 3% this year. High-income households are in a stronger position, buoyed by large property and stock market gains.

It is certainly true that higher energy prices may help to incentivise more investment in the US energy sector after the fourth-quarter 2025 GDP report again highlighted the bifurcation in corporate capital expenditure. Tech-related investment tied to software and computing power is growing 25% year-on-year, yet all other business capital expenditure has fallen in year-on-year terms for five straight quarters. This lack of breadth in the growth story is similarly seen in the jobs market, where, outside three sectors – leisure & hospitality, government and private education & healthcare services – the economy has lost more than 400,000 jobs since August 2024.

Fed rate cuts delayed, not cancelled

For now, we are forecasting GDP growth of 2.6% this year. The first quarter will be lifted by a rebound in federal government spending after the shutdown in the first six weeks of 4Q25, with strong tech-related investment and high-income consumer spending keeping the US economy moving along nicely in the second quarter.

However, the lack of breadth to the US growth story may become more of a vulnerability in the second half of the year. The longer energy costs stay elevated, the greater the risk it becomes demand destructive, which dampens inflation pressures over the medium to longer term.

The Fed will likely be nervous about headline inflation initially, but if the core metrics (excluding food and energy) start to cool, officials will likely become more comfortable cutting interest rates a couple of times in the second half of the year.

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